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The Future of Industrial Gas Turbines: Technological Advances and Market Trends

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Industrial gas turbines play a fundamental role in modern energy infrastructure, serving as key enablers of reliable power generation and industrial operations. With rising global energy demand and the imperative to reduce its environmental impact, these turbines are undergoing continuous innovation. This study explores major technological advancements, including novel material applications, aerodynamic refinements, improved combustion techniques, and the increasing role of digital technologies. Furthermore, the transition to alternative fuels and the integration of renewables are examined as pivotal developments. The analysis underscores the importance of ongoing enhancements in efficiency, emissions reduction, and operational flexibility to meet the evolving needs of the energy sector. In addition to technological progress, economic and regulatory factors shape the development and deployment of industrial gas turbines. Stricter emissions policies and the push for decarbonization drive research into hydrogen and synthetic fuels, while digital monitoring and predictive maintenance improve operational reliability and cost efficiency. The role of industrial gas turbines in hybrid energy systems, combining conventional and renewable sources, highlights their adaptability in ensuring grid stability. These combined innovations reinforce their relevance as a critical component of future energy solutions, enabling the transition toward more sustainable and resilient power generation.
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Keywords: energy systems, gas turbine heat transfer, industrial gas turbines, power generation technology, hydrogen energy

1 Introduction

Industrial gas turbines are crucial components in modern energy systems, playing a central role in electricity generation, mechanical energy provision, and various industrial processes. As global energy demand rises and the pressure to develop more environmentally friendly and efficient energy solutions increases, industrial gas turbines are poised for significant transformation.

This study explores the current developments and future trends shaping the future of industrial gas turbines. The significance of industrial gas turbines extends beyond their immediate applications in power generation. Their efficiency, fuel flexibility, and ability to provide a stable power and heat supply make them indispensable in industrial settings, such as chemical processing, oil and gas (O&G) production, and district heating. Moreover, their aforementioned adaptability to various fuel sources, including natural gas, hydrogen (H_2), and synthetic fuels, aligns them with ongoing efforts toward decarbonization and energy diversification.

Over the past century, gas turbines have evolved significantly, with advancements in materials science, aerodynamics, cooling techniques, and combustion technologies playing a pivotal role in enhancing their performance. This article provides a historical overview of these developments, highlighting key milestones that have shaped modern industrial gas turbines [1].

In addition to historical progress, technological advances continue to drive the efficiency and reliability of industrial gas turbines. These advancements include cycle improvements, such as the incorporation of recuperators and reheat cycles, which enhance thermal efficiency. Additionally, innovations in compressor and turbine design, including high-pressure ratios and optimized blade geometries, contribute to improved performance [2]. The role of additive manufacturing (AM) in producing intricate cooling structures and lightweight components is also explored as a transformative force in gas turbine engineering [3,4].

One of the most pressing challenges in industrial gas turbine development is achieving greater fuel flexibility. As the energy sector moves toward a more sustainable future, the ability to operate on alternative fuels, such as H_2 and biofuels, becomes increasingly important. This analysis examines recent progress in combustion technology that enables lower emissions and higher efficiency when using these alternative fuels [5].

The digitalization of industrial gas turbines has also emerged as a major trend, revolutionizing the way these machines are operated and maintained. Industry 4.0 concepts, including predictive maintenance [6–11] and digital twin technology [12–15], are reshaping maintenance strategies and enabling real-time performance optimization. By leveraging advanced data analytics and machine learning, operators can anticipate potential failures, reduce downtime, and optimize energy output.

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Furthermore, the integration of industrial gas turbines with renewable energy sources is gaining traction as a means to enhance grid stability and increase overall energy system resilience. Hybrid power plants, which combine gas turbines with renewable sources such as wind or solar, offer a promising approach to balancing intermittent generation and ensuring reliable power delivery [16,17]. Additionally, combined heat and power (CHP) systems continue to be a key application for industrial gas turbines, maximizing overall energy utilization by capturing and repurposing waste heat [18,19].

From a global perspective, industrial gas turbines play a vital role in meeting energy demand across diverse markets. This research explores regional trends in gas turbine deployment, including the impact of policy frameworks, market dynamics, and economic considerations on their adoption. The role of gas turbines in emerging economies, where demand for flexible and reliable energy solutions is growing rapidly, is also discussed [20–23].

Looking ahead, the future of industrial gas turbines will be shaped by both technological and market-driven factors. Key challenges include further reducing emissions, improving efficiency, and ensuring long-term fuel availability. The ongoing development of advanced materials, hybrid energy systems, and digital control technologies will play a crucial role in addressing these challenges [20].

2 Historical Development and Significance of Industrial Gas Turbines

The inception of gas turbine technology can be traced back to the late eighteenth and early nineteenth centuries. The earliest conceptualizations of the gas turbine engine emerged from the pioneering works of John Barber, who patented a design for a "Gas Turbine" in 1791. Barber's design, although primitive and not practical for power generation, laid the groundwork for future developments by outlining the basic principles of the gas turbine cycle [24].

In the late nineteenth century, engineers like Karl de Laval, Sir Charles Parsons, and George Brayton contributed significantly to the evolution of turbine technology. De Laval and Parsons, known for their development of the steam turbine, indirectly influenced gas turbine technology by demonstrating the potential of turbine-driven machinery. Meanwhile, Brayton's work on the Brayton cycle, a thermodynamic cycle that underpins gas turbine operation, provided a theoretical foundation that would later be realized in practical applications [25,26].

The early twentieth century saw significant advancements as the principles of gas turbines began to be translated into workable engines. In 1903, Norwegian engineer Ægidius Elling successfully built and operated the first gas turbine capable of producing more power than it consumed, albeit with limited efficiency. His work demonstrated the feasibility of gas turbines for practical use, marking a crucial milestone in their development [27]. Probably due to the fact that there was no real industrial basis in Norway at that time, his work went largely unnoticed, but that of the Armengauds in 1905 did attract attention (even though their gas turbine did not provide a net power output). This was also reflected in the literature of the time: Stodola's seminal work on turbomachinery (1905) was already entitled steam and gas turbines [28]. The main problem in these early years was manufacturing a compressor with a high enough efficiency, something that would come back and haunt gas turbine engineers in the decades to come. During the 1930s and 1940s, the pursuit of jet propulsion for aircraft provided a substantial impetus for gas turbine development. Frank Whittle in the United Kingdom and Hans Joachim Pabst von Ohain in Germany independently developed the first operational jet engines, proving the viability of gas turbines for high-speed aircraft propulsion [29]. The first jet engine to enter series production was the Jumo 004 in 1944. The success of these early jet engines highlighted the potential of gas turbines for other applications, including power generation and industrial use [30].

The post-World War II era saw a rapid expansion in the application of gas turbine technology beyond aviation. The need for efficient and reliable power generation solutions spurred innovation in industrial gas turbines. Advances in materials science, particularly the development of high-temperature alloys, enabled gas turbines to operate at higher temperatures and pressures, significantly improving their efficiency and power output [31].

In the 1950s and 1960s, heavy-duty gas turbines began to be widely adopted for electricity generation and mechanical drive applications. Companies like General Electric (GE), Siemens, BBC, and Westinghouse played pivotal roles in the commercialization and widespread adoption of gas turbines. These turbines were initially used as peaking power plants due to their ability to start up quickly and respond to demand fluctuations. However, as technology progressed, they increasingly became a staple for base-load power generation and combined cycle plants [1,32]. At the same time, small industrial gas turbines were developed, mainly based on aero-engine designs, for example, by Solar Turbines. Other companies manufacturing industrial turbines at the time were Snecma, Sulzer, and GEC Alstom. Kawasaki Heavy Industries (KHI) entered the market in the 1970s.

The development of combined cycle technology in the latter half of the twentieth century marked a significant evolution in gas turbine applications [33]. Combined cycle power plants (CCPPs) integrate gas turbines with steam turbines to utilize the waste heat from the gas turbine exhaust to generate steam, which in turn is used to operate a steam turbine and thus generate additional electricity. This integration greatly enhanced the overall efficiency of power plants, often achieving efficiencies exceeding 60% [25,26]. Combined cycle technology has become the dominant configuration for new large power plants due to its superior efficiency and lower emissions compared to conventional fossil fuel power plants. The ability to quickly ramp up and down also makes these plants ideal for complementing intermittent renewable energy sources like wind and solar power.

The applications for industrial gas turbines are mostly heat-driven, i.e., their purpose is to generate heat or cooling as well as electricity, the focus being on the thermal aspect. Consequently, their efficiency is measured as a combination of both, and as such, it can exceed 90% [34]. Industrial gas turbines can be used to generate hot water, steam, cooling, or a combination of two or even all three forms of heat.

In recent decades, advancements in computational fluid dynamics (CFD), materials engineering, and control systems have further propelled the performance and reliability of industrial gas turbines. Modern gas turbines are equipped with sophisticated control systems that optimize performance in real-time, reducing emissions and increasing fuel efficiency. The use of advanced ceramics and thermal barrier coatings allows turbines to operate at unprecedented temperatures, pushing the boundaries of efficiency even further [35,36].

The push toward decarbonization and sustainability is shaping the future trajectory of industrial gas turbines. Research is ongoing into H_2 combustion and carbon capture technologies, which hold the potential to significantly reduce the carbon footprint of gas turbine power generation. H_2, in particular, is seen as a promising fuel due to its zero-emission combustion profile. Companies are actively developing gas turbines capable of operating on 100% H_2 or H_2–natural gas blends, aiming to make them viable solutions for a low-carbon energy future [37–39].

3 Technological Advances

Building upon the historical perspective of industrial gas turbines, the subsequent sections delve into the technological advancements that will shape their development. This section explores key innovations and developments that have enhanced the performance, efficiency, and adaptability of gas turbines in modern energy systems.

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The discussion begins with Sec. 3.1, highlighting improvements in thermodynamic cycles and compressor technologies. This is followed by Sec. 3.2, with dedicated Secs. 3.2.1–3.2.3, which are critical for managing the high thermal loads in modern turbines.

Next, Sec. 3.3 is examined, detailing adaptations that allow turbines to operate efficiently with a broader range of fuels. The section then transitions to Sec. 3.4, showcasing the transformative impact of digital technologies. This section includes Sec. 3.4.1, emphasizing condition monitoring and fault detection, and Sec. 3.4.2, which illustrates the integration of virtual modeling and remote capabilities.

The section concludes with Sec. 3.5, discussing the role of gas turbines in supporting sustainable energy systems. This includes Sec. 3.5.1, which showcases potential optimizations of energy utilization, and Sec. 3.5.2, combining gas turbines with renewable energy sources for enhanced flexibility and reliability.

Through these topics, the following sections provide a comprehensive overview of how cutting-edge innovations are driving the continued relevance and advancement of industrial gas turbines in a rapidly changing energy landscape.

3.1 Cycle Considerations and Advances in Compressor Design. This article focuses on stationary gas turbines with a power output up to 50 MW. Unlike large-scale heavy-duty gas turbines that are nearly exclusively installed as prime movers in large-scale CCPP gas turbines in the considered range are mostly used in heat-driven CHP plants, or may provide residual services as flexible peakers in simple cycle operation [21]. In the considered power range, they may compete with gas engines in CoGen-plants [40]. A significant number of industrial gas turbines are moreover used as mechanical drives for compressors and pumps in, e.g., oil and gas industries [22]. This usage is not in focus here, and it is safe to say that technological trends identified in relation to the core engine are applied regardless of the usage of the engine.

Consequently, the design principles of small and medium-sized gas turbines differ from those of large-scale ones, in so far as greater flexibility with regard to fuels and a wide operating range are required. In addition to this, small and medium plants are offered to a wider range of customers stretching from local utilities, private companies, and the service sector to guarantee a decentralized supply of heat and electricity in, e.g., hospitals or large housing and office districts. Therefore, the design must accommodate requirements of both low capital expenditures and operational expenditures, and meet the requirements of a robust design to guarantee a high level of availability at moderate maintenance efforts.

For mechanical drive applications, the preferred configuration is a two-shaft design, in which the power turbine is mounted as a separate unit from the gas generator. This setup allows flexible operation of both shafts, as the speed of the power turbine is decoupled from the compressor speed. Some original equipment manufacturers (OEMs) offer both single-shaft and two-shaft gas turbines of the same series where the gas generator is built of identical compressor and combustor sections with slight modifications in the high-pressure turbine stages. For the two-shaft design, the last turbine stages of the single-shaft engines are replaced by freely spinning power turbines that are only aerodynamically coupled with the gas generator [41,42].

The efficiency of the ideal Brayton Cycle with isentropic compression, expansion, and isobaric combustion is determined solely by the pressure ratio. In practice, this is also coupled with the turbine inlet temperature to maximize the specific work of the gas turbine at the target value of the turbine exit temperature that determines the performance of the bottoming cycle.

Figure 1 provides an overview of the trends for industrial gas turbines offered by the major gas turbine producers worldwide [22]. Overall, the pressure ratio increases with the shaft power and ranges between 15 and 20 for gas turbines under 20 MW, and reaches 30 for larger shaft powers. In addition to the traditional industrial gas turbines, a second group of gas turbines is added to

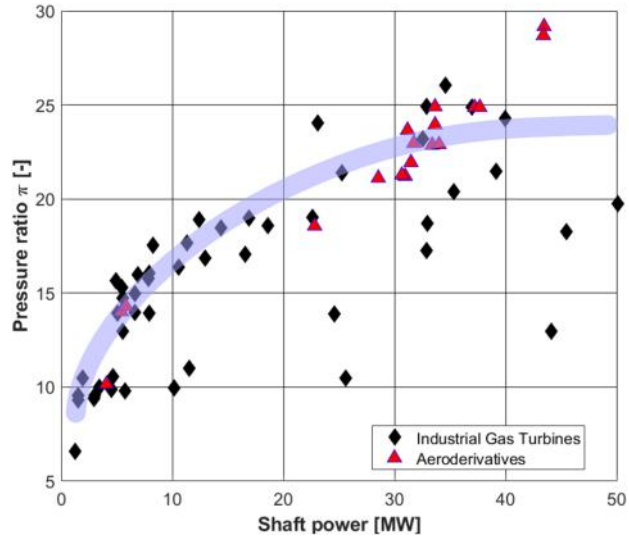

Fig. 1 Pressure ratios of industrial gas turbines and aeroderivatives of major OEMs up to 50 MW

the graph that is derived directly from aero-engine core engines with only slight modifications, which are referred to as aeroderivatives [43]. As shown in Fig. 1, they are characterized by high thermal efficiencies and are only excelled by industrial gas turbines of the most recent generation.

To quantify the technology level of industrial gas turbines, the efficiency is plotted versus the cycle pressure ratio, and the slope is compared to the overall potential of the ideal Brayton cycle (Fig. 2). Here, a clear tendency can be observed across the wide range of pressure ratios in existing gas turbines. In fact, the trend closely follows a graph that approximately aligns with the line defined by 70% of the ideal pressure ratio of the Brayton cycle. This indicates that technological barriers common to all gas turbine producers are clearly identified. Overcoming these barriers requires considerable effort to achieve technological advancements, which will be outlined in this contribution. Similarly, aeroderivative engines achieve high efficiency levels with designs derived from aero-engine technologies of the late 1980s and 1990s [43].

If gas turbines are used in CCPP and CHP plants, large values of the turbine outlet temperature are required to achieve the optimum level of fuel exploitation.

Figure 3 shows the importance of the exhaust gas enthalpy for CHP efficiency: some engines that have a high electrical efficiency (e.g., engine 2) have a fairly low overall CHP efficiency because the high electrical efficiency entails low exhaust gas temperatures (Fig. 4). On the other hand, an engine with a low electrical efficiency (e.g., Engine 6), may have a very high overall efficiency. This is because a low electrical (thermodynamic) efficiency causes a high exhaust gas temperature [41,42].

If turbine exit temperatures above 500 °C are considered suitable for exhaust heat recovery plants, it becomes evident that most aeroderivative turbines are less efficient compared to industrial gas turbines optimized for combined applications (Fig. 5). The reason is that aeroderivatives combine high-pressure ratios with moderately high turbine inlet temperatures (TIT), leading to low turbine outlet temperatures. Nevertheless, aeroderivatives bear advantages in being more compact and having less weight, which are important for various installations, for example, mobile electricity units [22]. Smaller gas turbines have a higher cost per megawatt than larger engines because scaling allows OEMs to reduce manufacturing costs (per kilowatt) as unit power ratings and physical size increase [44]. Hence, it is more difficult to incorporate advanced cooling and sealing technologies in small engines. On the other hand, smaller gas turbines perform well because the limited

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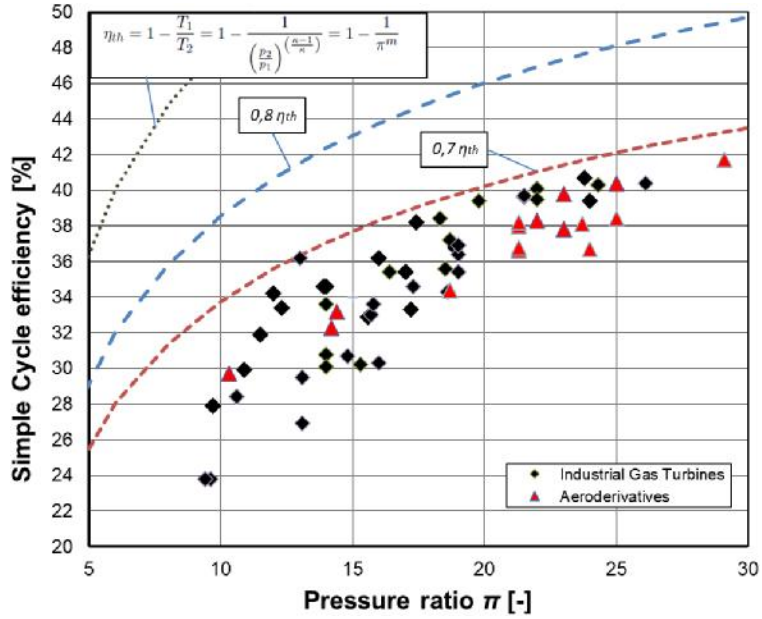

Fig. 2 Simple cycle efficiencies (referred to the lower heat value of fuel) of industrial gas turbines and aeroderivatives in the power range up to 50 MW

pressure rise is associated with relatively low TIT, not demanding excessive cooling consumption. Recuperated gas turbines, for example, can only achieve high efficiencies at pressure ratios of around 10 [45]. The technology level of small-sized gas turbines was very often curtailed by cost considerations in the past, but is now driven forward due to harsh competition among the OEMs.

Following the previous discussion that linked the compressor pressure ratio with overall efficiency, it is obvious that advances in gas turbine compressor design are important. Moreover, to achieve high CHP performance, the turbine inlet temperature has to be raised along with the pressure ratio, with consequences for the material and cooling requirements of modern expansion tracts. This will be discussed in detail in the following sections.

As compressor performance (pressure ratio) increases, both efficiency and compressor stability need to be considered. In particular, if the compressor shall be operated at a wide operating range and with enhanced fuel flexibility, new, dedicated, technologies need to be introduced to compressor design standards that have up to now been developed in parallel with aero-engine compressors [46]. This follows a long-standing tradition in which many OEMs directly adopted aero-engine technologies in the design of industrial gas turbine compressors based on cooperation with aero-engine producers [47].

Figure 6 outlines the state of performance of a number of compressors used in current industrial gas turbines. To compare the designs, all data are scaled to the size of a 10 MW gas turbine using a modified Reynolds-number approach. In addition to the adiabatic efficiency, defined as the quotient of ideal to real total enthalpy differences, lines of constant polytropic efficiencies are drawn in the graph. The relationship between adiabatic efficiency η_{ad} and polytropic efficiency η_{pol} is given by

$$\eta_{ad} = \frac{\pi^{\frac{\gamma-1}{\gamma}} - 1}{\pi^{\frac{\gamma-1}{\gamma\eta_{pol}}} - 1}$$

with total pressure ratio π and ratio of specific heats γ.

The use of the polytropic efficiency allows comparison of technology standards of compressors at a very wide range of pressure ratios. It is shown that currently, for the considered size scale, the line with a polytropic efficiency of 91% forms the barrier to most up-to-date designs.

Advances in high-speed compressor design have led to a considerable increase in stage loading and stage efficiencies. As shown in Fig. 7, the average stage pressure ratios of 1.25 characterize the best performing compressors of recently developed industrial gas

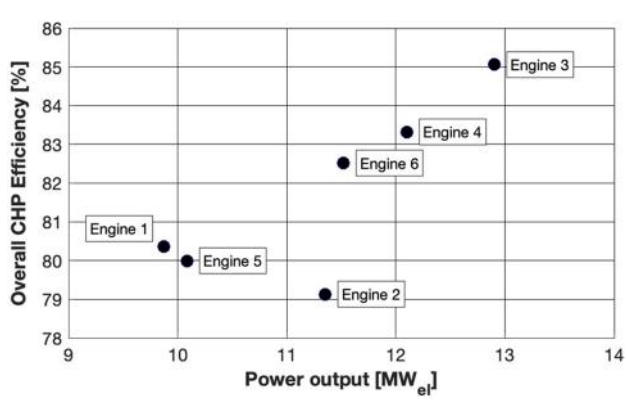

Fig. 3 Overall CHP efficiency (referred to the lower heat value of fuel)

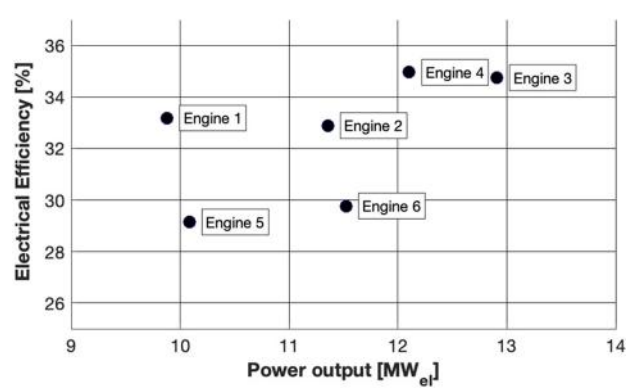

Fig. 4 Electrical efficiency (referred to the lower heat value of fuel)

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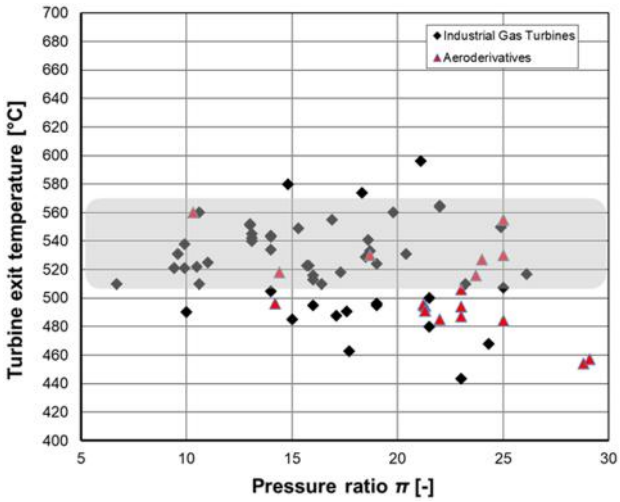

Fig. 5 Turbine exit temperatures of industrial gas turbines and aeroderivatives in the power range up to 50 MW

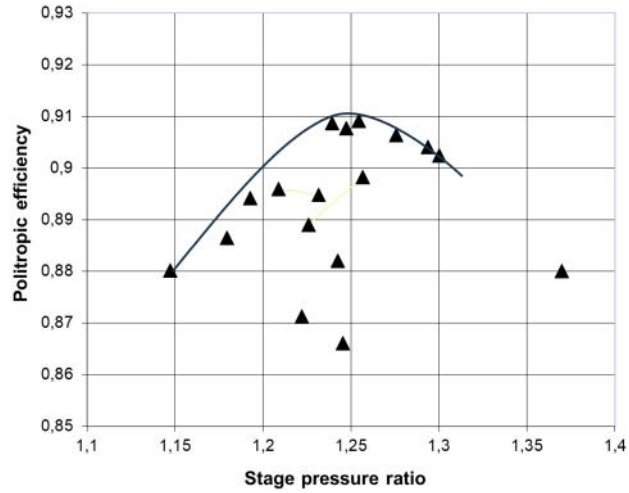

Fig. 7 Technological trends of mean stage pressure ratios (scaled to 10 MW size gas turbine)

turbines. This observation demonstrates the high technology level in blade and vane design at transonic speeds. As a result of the higher possible pressure ratio, the number of stages has been reduced by about 30–40% over the last two decades, leading to more compact designs, a considerable decrease in wetted surfaces in the compressor, and savings in weight, materials, and cost.

Gas turbine plants embedded in energy systems dominated by volatile renewable energy sources need to operate more flexibly with respect to load changes and fuel variability. Consequently, compressors must be designed to ensure safe operation over a wider range and to be more resistant to flow separation [2]. The stability limits can be extended by application of casing treatment (CT). However, optimization of CT is linked to the individual design of the airfoils and cannot be generalized [48,49]. Moreover, because fouling may diminish the effect of CT after a relatively short operating time, it is currently not considered in industrial gas turbines.

In general, gas turbines in the considered range may face harsh operational and environmental conditions, in particular at chemical plants or in offshore installations of the oil and gas industry [50]. Though the inlet of gas turbines is protected by elaborate filter cartridge systems [51], the front stages of the compressor may be exposed to foreign object damage (FOD) [52] due to polluted intake air, causing either erosion or damage on the blade surfaces that would affect the compressor performance and stability [53]. Frequent washing of the compressor flow path is a common precaution against compressor fouling to restore the performance [54]. Icing is a phenomenon not only restricted to aero-engine compressors but also occurring in the front stages of industrial compressors under certain climate conditions, and may lead to FOD in the front stages. Its mechanism needs to be understood, and anti-ice measures are required, e.g., discussed by Assady and Sammak [55].

The development of active and passive surge control systems has gained increased attention and covers a wide variety of solutions. For details, see the review articles by Alsuwian et al. [56] and Blanchini and Giannattasio [57].

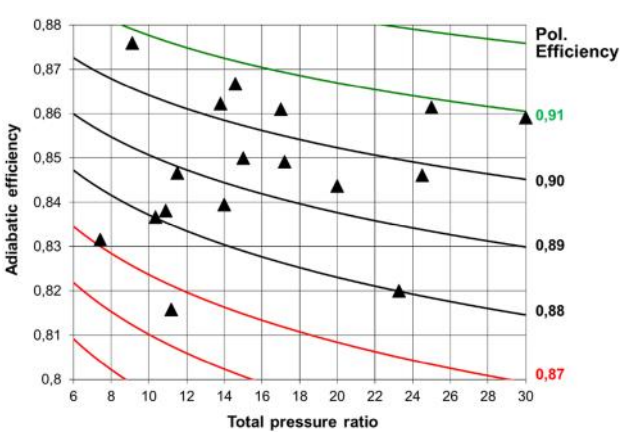

Fig. 6 Technological trends of gas turbine compressors (scaled to a 10 MW size gas turbine)

3.2 Enhanced Turbine Design. The following subsections examine the multifaceted advancements in turbine technology. First, key material innovations, including cutting-edge manufacturing techniques, are explored. Subsequently, emphasis is placed on aerodynamic refinements and enhanced thermal management strategies to optimize turbine performance.

3.2.1 Materials Science. New alloys and coatings that withstand higher temperatures contribute significantly to efficiency improvements. Examples include nickel-based superalloys [58] and ceramic matrix composites (CMCs), which remain stable under extreme temperatures while maintaining high strength.

Even though CMCs can withstand very high temperatures, their geometric adaptability is fundamentally limited. This means it is very difficult and often impossible to integrate sophisticated cooling schemes into CMC parts, offsetting much of their high-temperature capabilities and limiting their use to geometrically simple components. Some efforts have been made to investigate their suitability for film cooling [59].

The evolution of superalloys now seems to be geared toward novel production methods, i.e., AM and the development of alloys suitable for AM. Numerous AM methods exist, but by far the most important for gas turbines are direct energy deposition and laser powder bed fusion, which is also called selective laser melting (SLM). The former is used especially for repair, as it does not rely on a powder bed. This is of relevance for service parts, as repair is generally cheaper and quicker (for legacy engines). The relative simplicity of the cooling schemes makes the economic argument more difficult, though. The latter is primarily used to manufacture new, high-temperature components [3]. AM via SLM essentially consists of numerous very thin welds, making the utilization of non-weldable materials (i.e., most nickel-based alloys) not suitable for AM. Although cobalt-based materials like MAR-M509 are suitable, their prohibitive costs have seen

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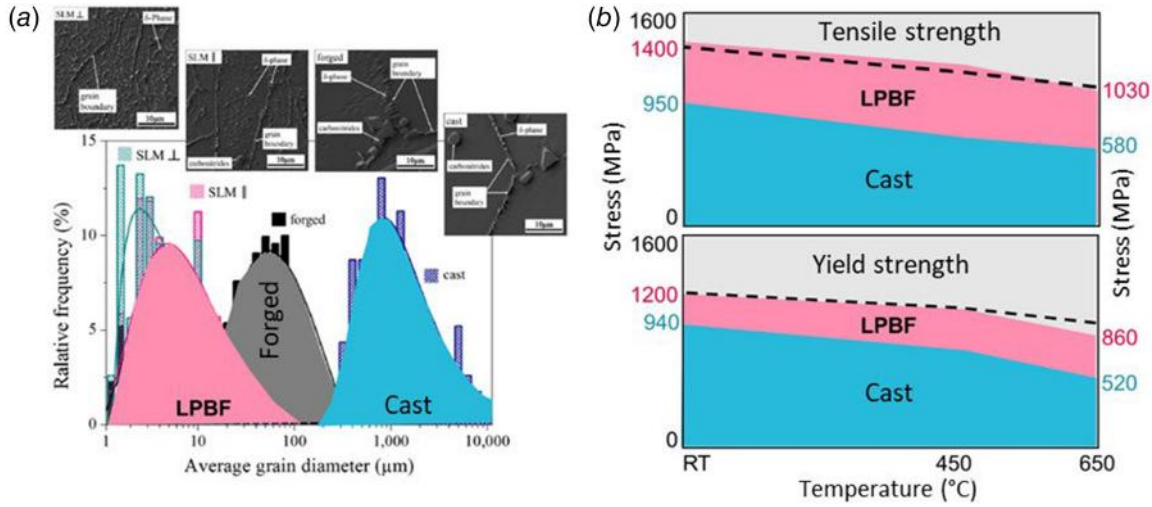

Fig. 8 Comparison of material properties: (*a*) grain sizes, based on Ref. [60] and (*b*) mechanical properties structure

them used less rather than more as engine alloys over the last decade. The material currently used most often for AM is Inconel 718, as this has high iron content. Unfortunately, the material properties of Inconel 718 at high temperature are not as good as those of most other nickel-based alloys.

Still, the properties are better than cast and worse than forged samples of the alloy (Fig. 8) @ Ref. [60]. Compared to forged Inconel 718, the creep behavior and notch-creep embrittlement of AM samples need to be better understood, as the established heat treatments to reach forging quality do not work commensurately with AM material [61]. This may be caused by porosity and surface roughness, which give ample opportunity for crack initiation and growth, as well as residual stress, which also causes cracking. Stresses can also cause warping, as illustrated in Fig. 9, which shows a preliminary design for a cooling insert. Therefore, much time has been spent on the development of materials that are both suitable for AM and have the required material properties. One of these is ABD900® [4,60,62–64], but other, lower-temperature materials based on titanium or aluminum are under development as well, for example, AlSi10Mg and Scalmalloy. Various OEMs have reported high-temperature parts being produced by AM, either for series or test engines. Among them are GE (fuel nozzles), Siemens (blades; a particular and dedicated emphasis on vanes [65]), and MAN (nozzle guide vanes and blades, see Fig. 10). This figure also illustrates that the surface quality is excellent and the deviations to design intent minimal (or at least not larger than with investment casting). In fact, the effect of surface roughness is relatively small for AM channels with the diameters of legacy designs.

These manufacturing capabilities together with the suitable materials have the potential of fundamentally changing both the design of high-temperature components as well as maintenance. The first is obvious and often investigated [66]: significantly smaller cooling features can be created with higher relative positional accuracy than with investment casting [67]. This, in turn, means the cooling can be more effectively adapted to the outer

Fig. 9 Warped preliminary (non-optimized) design of an impingement insert for an nozzle guide vane (including the supporting structure)

Fig. 10 Internally cooled turbine rotor blade (SLM-manufactured)

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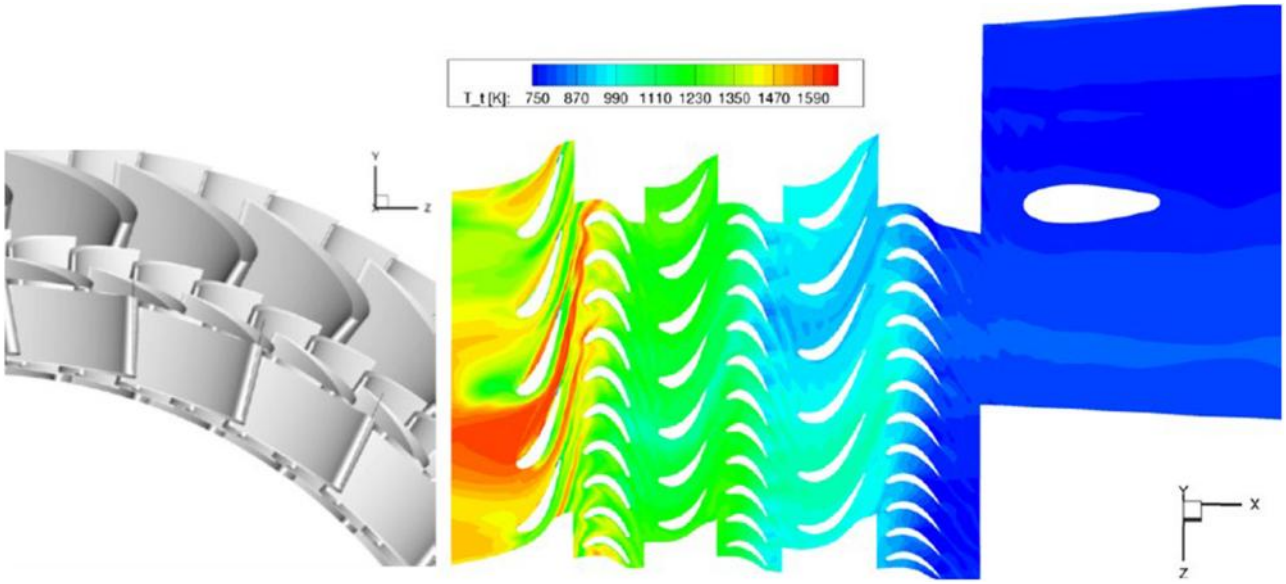

Fig. 11 Visualization of combustor–turbine interaction processes

heat loads on the cooled component, with greater flexibility. It should be noted, though, that channels the size of standard film-cooling holes remain extremely difficult to manufacture additively with precision. Therefore, such holes are, in practice, most of the time not printed at all, but are created using electrical discharge machining or laser drilling [68]. The latter could also be used to improve printed geometries, provided that the positioning of the laser relative to existing film-cooling holes is precise enough.

The flexibility provided for maintenance should not be underestimated either: AM will enable the reduction of costly stocks of replacement parts, especially hot gas components and airfoils. The flexibility of manufacturing legacy parts "as is," or replacement parts with enhanced cooling capabilities is significant. The design of parts with enhanced cooling is more challenging because, first of all, the often-repeated effect of surface roughness for small cooling holes and intricate AM cooling designs comes into play. Second, CFD tools have not been validated for these designs, meaning that larger deviations between analytical (if any), numerical, and experimental results must be expected.

3.2.2 Aerodynamic Optimization. Continuous improvements in turbine blade design optimize airflow and minimize losses. CFD and other advanced simulation methods enable engineers to develop designs that ensure maximum energy conversion [69]. As turbines are exposed to very high gas temperatures, the design of gas turbine blades and vanes demands for highly interdisciplinary design systems covering thermal and structural design considerations. To accommodate internal flow paths for the coolant inside the airfoils, the leading edge thickness needs to be increased to allow for the ejection of the cooling flow (as showerhead or film-cooling) into the primary flow path, causing additional losses and exit flow angle deviations at transonic outlet Mach numbers (base-pressure losses, see Ref. [70]). Unlike flows in axial compressors, turbine flow fields are more strongly dominated by the occurrence of secondary flows due to high turning and stage pressure ratios that strongly interact with heat transfer properties along the side walls [71]. Moreover, secondary flow phenomena must be thoroughly understood, as they not only contribute to additional aerodynamic losses but also influence the local development of the cooling film along the wetted surfaces of film-cooled blades and vanes [72]. To reduce secondary losses, contoured endwalls—either axisymmetric or non-axisymmetric—are considered. Normally, this is achieved through an axisymmetric reduction of the span between the leading edge and trailing edge. This approach decreases flow velocity in the high-turning upstream part of the vane, thus mitigating the emergence of passage crossflow. Additionally, the passage contraction induces an acceleration that thins the endwall boundary layer in the downstream region, further reducing secondary flow. Non-axisymmetric contouring stretches from the suction to the pressure side of the profiles and mainly controls the secondary flow pattern locally, and the optimization of the endwall shape has been an essential matter of research [73–75]. Moreover, its manufacturing is more elaborate and expensive, and, therefore, due to economic restrictions not the desired option for small and medium size industrial gas turbines. Instead, more focus is on profile shaping itself to reduce secondary flows by influencing the loading distribution in the streamwise direction to improve both aerodynamic performance and combustor–stator interaction [76–78].

With regard to the stage matching, 3D multistage-CFD has become a key tool since Denton first published a multistage Euler solver for turbomachinery flows [79]. With the development of powerful computer capacities, unsteady CFD computations have become the standard for the design phase of turbines. Fully 3D transient flow-field analysis has become feasible for detailed optimization at modern design departments, even for multi-stage turbomachinery [80]. Figure 11 shows the example of the interaction between the combustor and a three-stage turbine for a 10 MW-class gas turbine computed with a customized unsteady Reynolds-averaged Navier–Stokes (URANS) solver based on Denton's TBLOCK. With six combustor cans in total, a 60 deg sector of the turbine with adjusted blade and vane counts was modeled to get insight into the propagation of the hot streak leaving one of the combustors. Based on these results, the temperature load on the stator vanes can be identified, and the cooling schemes of each stator vane could be independently adjusted to reduce overall coolant consumption. Furthermore, the optimum location of the turbine exhaust diffuser struts and temperature probes at the turbine exit plane that are used for power control of the turbine could be projected (Fig. 12).

For these computations, results of which are shown in Figs. 11 and 12, the combustor outlet condition was assumed to be steady-state, which does not conform to reality. To obtain more insight into details, a more comprehensive model was set up by Lo Presti et al. [81] using detached eddy simulation (DES) turbulence modeling. The analysis was run with a full 360 deg sector with real-blade counts and gives detailed results of the transient behavior of the flow field. Using DES or large eddy simulation

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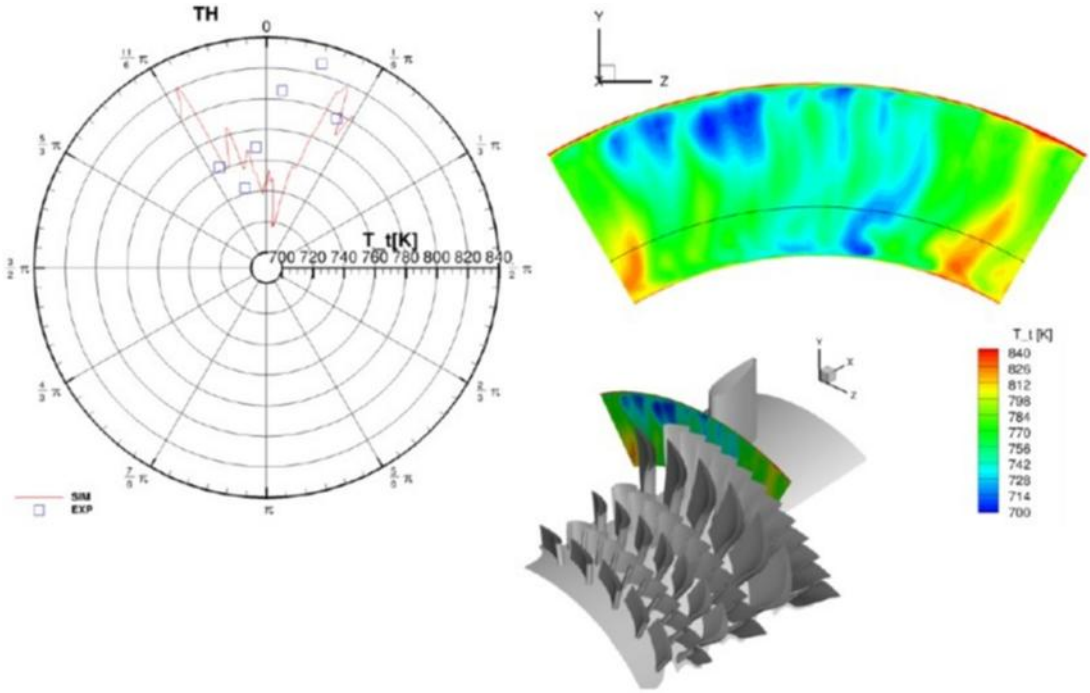

Fig. 12 Computation of the combustor–turbine interaction of a single-shaft three-stage MAN turbine and comparison with experiments at the turbine exit plane

modeling allows to identify and mitigate unsteady losses (Fig. 13) @ Ref. [81].

Advanced turbulence modeling supports the design of exhaust diffusers of gas turbines. As the outlet Mach number and flow angle deviation from the axial direction can be of significant magnitude at off-design conditions, the potential of diffuser losses is not negligible for the overall cycle efficiency. Figure 14 shows the differences of computed vorticity lines with both a conventional URANS model and a scale-adaptive turbulence simulation–shear stress transport model (SAS–SST) [82]. Unlike the conventional turbulence model application of the SAS–SST model, this implementation suppresses flow separation due to the strong turbulent mixing, which is in agreement with model experiments where the rotating blades are only modeled by spikes [83]. Recently, new correlations were developed on the basis of these experiments for the practical design of gas turbine diffusers [84].

3.2.3 Cooling Systems. Increasing the TIT in line with the pressure ratio increases the thermodynamic efficiency toward the optimum value of the ideal Brayton cycle, which is determined by the pressure ratio as discussed in Sec. 3.1. Temperatures have long exceeded those at which metallic alloys can operate over long periods of time. Therefore, hot gas path components need to be made of better materials, cooled or shielded from the hot gas. The coolant represents a thermodynamic loss and is only used to cool components that would otherwise overheat. Therefore, reducing the amount of coolant used by improving materials is essential.

Advances in turbine blade cooling enable higher operating temperatures without compromising structural integrity. Modern cooling systems use internal channels and advanced coatings to efficiently cool the blades, extending their lifespan [67]. Currently, the amount of coolant used is between 20% and 33% for modern stationary gas turbines. Naturally, small industrial gas turbines and, above all, older models with significantly lower turbine inlet temperatures tend toward the lower end of this span. While there is a need to develop new cooling strategies with new engine designs, the requirements for legacy engines should not be underestimated either. As long as a new cooling design for such an engine does not necessitate an overhaul of the SAS, reducing the coolant in the blades and vanes can provide both thermodynamic and financial benefits.

The classical cooling schemes consist of impingement cooling [85,86] and pin-fin cooling [87,88] for the vanes, and rib-turbulated cooling [89–91] and pin-fin cooling for the blades. Both blades and vanes in the first stages normally feature film cooling [92,93], in addition to other cooling techniques [89]. This setup has not fundamentally changed over the last decades (see Ref. [94] for an overview of a typical cooling setup) and, consequently, holds limited potential for improvement, i.e., a reduction of component temperatures. Furthermore, the much smaller dimensions of industrial gas turbines make it more difficult to incorporate cooling designs from large engines into the blades and vanes at an acceptable cost and with an acceptable precision. Therefore, the design choices for small industrial gas turbines may differ from those for larger gas turbines. Aero-engines suffer less from this problem due to the significantly higher number of engines built and the higher acceptable costs per engine.

Over the last few years, there has been some interest in using "phantom cooling," which can be used equally well for large and small gas turbines. This means the effect of purge and leakage flows on the surface temperature of components is quantified, and the active cooling is reduced accordingly [95–99]. As can be seen in Fig. 15, Ref. [95], the purge air emitted from a slot between the combustion chamber and the nozzle guide vane, which is necessary for the assembly of the engine, contributes significantly to the cooling of the endwalls. This effect is markedly different for old and new aerodynamic stator designs.

As Fig. 16 @ Ref. [100] shows, the designs of the slot or cooling holes have a significant impact on the cooling distribution as well as on the secondary flow. Although such modifications will certainly help to reduce coolant consumption in legacy engines, they will probably not be enough to significantly impact the design of new engines. For this reason, some novel concepts have garnered attention recently, all of which are by and large, enabled by AM [101].

Effusion cooling has long been used in combustion chambers and transition ducts, and has seen some limited use in other hot gas sections [102]. Some aspects of effusion cooling have gained

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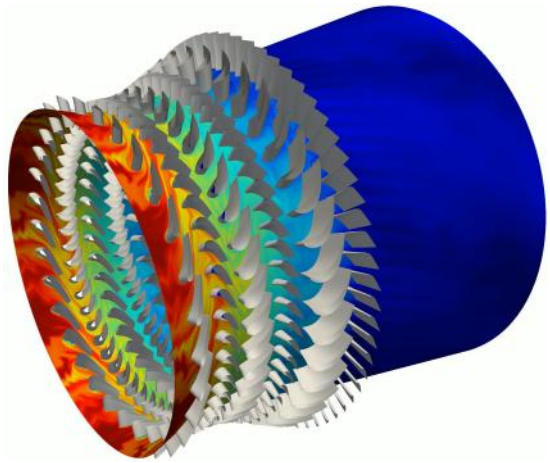

Fig. 13 Computation of the combustor–turbine interaction of a single-shaft three-stage turbine with a DES scheme in a 360 deg-model with real-blade counts [81]

traction with the advent of AM, in particular double-wall cooling, i.e., the combination of impingement and effusion cooling. Manufacturing these geometries has become much easier using AM. Older investigations were carried out at Yonsei University in Korea [103], whereas the topic is currently pursued predominantly at the University of Oxford (UK) [104]. Studies for AM airfoils for small industrial gas turbines have been reported in Ref. [105]. The vane tested in this study featured a double-wall cooling design incorporating micro-channels, which demonstrated slightly improved performance compared to the baseline designs: impingement cooling for vanes and serpentine cooling for blades.

The utilization of H_2 and its derivatives (e.g., ammonia (NH_3) and methanol (CH_4O)) will also have an impact on cooling design. The main reasons are the changes in radiative and convective heat transfer caused by the changes in species in the exhaust gas. Carbon dioxide (CO_2) will be completely eliminated, but the amount of steam will increase significantly. Estimates show that an increase in cooling efficiency of approximately 20% is needed to counteract this [106]. One of the biggest challenges is achieving fuel flexibility: attaining lean premixed combustion across a range of H_2 content from 0% to 100% is difficult, and NH_3 and CH_4O (which is liquid by nature) require entirely different burners. The resulting differences in exhaust gas compositions, as well as the vastly different turbine outlet temperature profiles, will have an undefined yet significant impact on the cooling design.

The impact of both artificial intelligence (AI) and machine-based learning on cooling design needs to be taken into account as well. Model-based learning was used for the adaptation of turbulence models for coolant–mainstream interaction, but the significant impact of the learning dataset precluded its use [107–110]. Other uses include the generation of load profiles for combustion chamber liners that can be equally well used for the evaluation of airfoils. Some advances were made using probabilistic methods in the design of SAS [111–113], and it remains to be seen how this could be extended to cooling designs. All of these advances do not only enable a more detailed assessment of the thermal load during steady-state operation (design point), but especially a better assessment of the impacts of part load operation and transients on the lifetime of the components.

Currently, there appears to be a reluctance to implement highly innovative—and therefore potentially costly—cooling concepts in small industrial gas turbines. This hesitation is primarily attributed to spatial constraints, which increase design complexity, and to uncertainties regarding future operational requirements and associated return-on-investment considerations. Nevertheless, OEMs

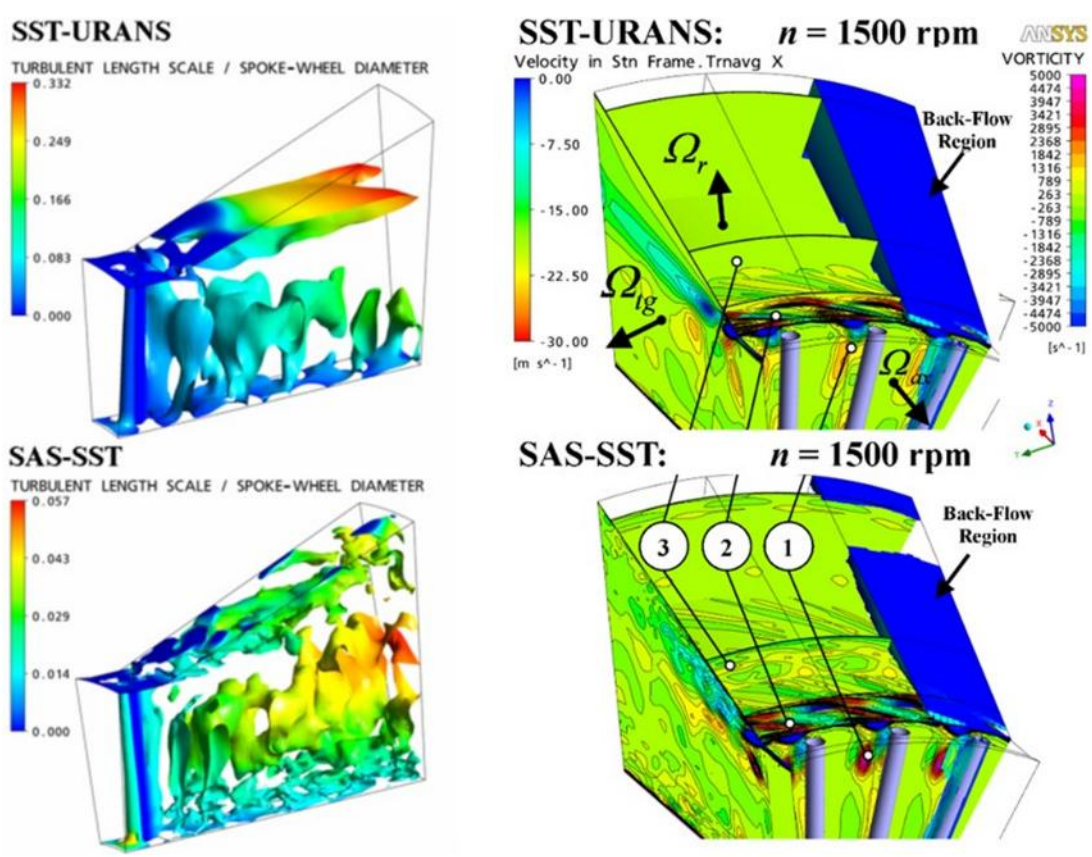

Fig. 14 Vorticity pattern in a gas turbine diffuser: impact of advanced CFD modeling on the prediction of diffuser pressure recovery (SAS–SST and SST–URANS)

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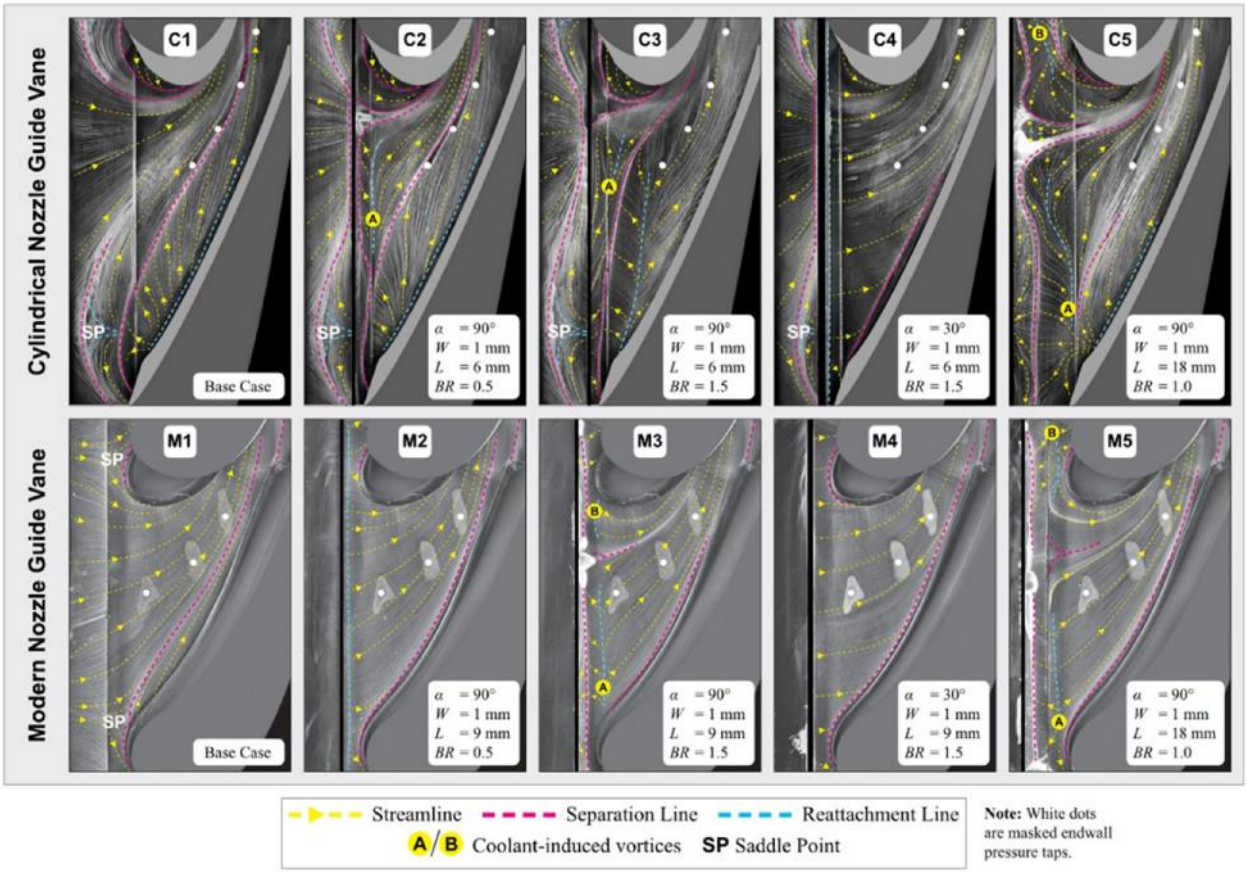

Fig. 15 Cooling effect of purge airflow on endwalls (top: legacy design and bottom: modern design) [95]

will need to engage with these advanced technologies, given their significant potential for cost reduction and the anticipated high cost of green fuels.

3.3 Enhanced Fuel Flexibility. Until very recently, the design principles of gas turbine combustors have been guided by the reduction of avoidable emissions inherent with the combustion of natural gas to meet legal requirements and to protect human health. All major gas turbine manufacturers have developed dry low-emission (DLE) technologies to cut down emissions of nitrogen oxide (NO_X) and carbon monoxide without curtailing the power output and efficiency of their gas turbines. As they are not in the main focus of this contribution, the reader is referred to the overviews in Refs. [5,114].

Apart from these advancements, fuel flexibility has become a driving factor in combustor development [115]. In particular, industrial gas turbines have traditionally been expected to handle a wide range of fuels because of their installation as mechanical drives in upstream and midstream oil and gas applications with very changeable qualities of the natural gas. Liquefied petroleum gases need to be considered in certain industrial power generation plants to use waste gases of technical processes (e.g., blast furnace gas). Moreover, industrial gas turbines of utilities in remote areas require the capability to combust liquid fuels in certain island applications. Another example is dual-fuel flexibility to allow both the burning of gaseous and liquid fuels in the combustor.

In the future, this flexibility will be much more important. When it comes to the avoidance of greenhouse gases, a fuel shift from fossil to green fuels, i.e., sustainable fuels derived from renewables, is the only promising way forward. Currently, (green) H_2 is the most prominent solution for power generation combined with inter-seasonal renewable energy storage of H_2 in liquid or gaseous form and its derivatives, such as CH_4O, NH_3, methane (CH_4), and synthetic fuels [116]. Biofuels are a complementary option and can currently be manufactured at a high, almost sulfur-free quality [117], but cannot cover the same (large) scale. Biofuels might be more suitable to be use in the mobility sector.

Gas engines and gas turbines can be converted from natural gas to H_2 operation [20]. Blending H_2 with natural gas and the conversion of gas turbines have been investigated and already demonstrated up to 100% H_2 [118]. The competitiveness of actual H_2-fueled gas turbines is driven by emissions restrictions when using different mixtures of H_2, natural gas, and biomethane. The highest competitiveness of H_2-fueled gas turbines (together with H_2 storage) is found in electricity systems with high shares of wind power [119], although their overall economic feasibility remains a matter of some debate [120]. When 100% H_2 is being used as fuel, the thermal efficiency of gas turbines is similar to that when using carbon-containing fuels.

Research is on the way to develop low NO_x combustors for a blend of natural gas with H_2 up to 100% by all major OEMs (e.g., Refs. [121–125]). Retrofitting of existing gas turbine plants without violating emission restrictions in NO_x is ready for the market as long as the amount of H_2 is less than 60% (by volume). For larger shares of H_2, wet low-emission combustors with injection of steam are successful options that are mainly used in aeroderivative gas turbines. However, the adjustment of dry low NO_x combustion chambers is at the foreground of the development. To avoid increased NO_x emissions and flashback inherent with the high flame speed of H_2-dominated fuel blends, fundamental changes in the combustion chamber need to be devised. The application of AM techniques allows for shaping complex multi-nozzle structures to encounter these problems and is in a mature development state by all major OEMs [126]. First full-scale installations have been demonstrated in certain publicly

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funded applications (HYFLEXPOWER [127], Flex4H2 [128], etc.).

The integration of NH_3 into gas turbines not only supports the decarbonization of the energy sector but also enhances energy storage and transportation capabilities [129]. Because of the low energy content per volume of H_2 as well as the problems storing and transporting it, NH_3 is favored as an energy carrier. All necessary technologies for the NH_3 value chain are known, as it is already the largest internationally traded chemical commodity. Combustion (unlike the use in fuel cell vehicles) does not require absolute purification of the fuel.

Although H_2 is in the focus of development in Europe, a significant amount of research is being conducted on NH_3 combustion in Japan [130]. The main drawbacks of NH_3 combustion are its low energy content (poor combustibility) and the production of NO_x as a byproduct. Furthermore, the combustion volume has to increase dramatically. The formation of NO_x requires some form of SCR, but, naturally, the NH_3 needed is readily available. These mixtures require approximately the same amount of coolant as pure hydrogen.

Applications involving 100% NH_3 gas combustion have been explored [131,132], but challenges include difficulties in igniting NH_3 gas and the need for a more easily ignitable fuel during startup. As was recognized very early, this can be mitigated by cracking the NH_3 nitrogen and H_2 and burning a mixture of 70–90% NH_3 and 10–30% H_2 [133,134]. A demonstrator for an NH_3/natural gas co-fired gas turbine was shown [135], revealing that NH_3 combustion generates significant quantities of nitrogen oxides through both thermal mechanisms and from the fuel itself. These emissions, along with unburned NH_3, need to be mitigated. Blending NH_3 with other fuels can improve emissions performance [136].

Park et al. examined whether co-firing H_2 and NH_3 with natural gas in an industrial gas turbine could enhance the environmental performance of existing combined cycle power plants [137]. Nose et al. [138] described the research and future plans for an NH_3 cracking gas turbine system, which burns H_2 decomposed from NH_3 using the exhaust heat of the gas turbine. They proposed two configurations: a co-firing system that burns mixtures of natural gas and cracked NH_3 gas, and a mono-firing system that burns only cracked NH_3 gas [138]. Nitrogen as ballast in the combustion gas rather simplifies the combustor design [139].

A burner capable of burning both kerosene and NH_3 was developed by Iki [140]. Initial operation with kerosene, followed by the gradual addition of NH_3—as already stated—resulted in high NO_x concentrations in the exhaust gases, which could be significantly reduced through selective catalytic reduction. Some other areas may be relevant, but have received much less attention. Compared to natural gas combustion, which involves various hydrocarbons, CH_4–air gas turbines use pure CH_4, resulting in cleaner and more efficient combustion [141]. For chemically recuperated gas turbines (CRGT) based on CH_4 steam reforming, the impact of various operating parameters on the efficiency of the CRGT was determined [142,143].

3.4 Digitalization and Industry 4.0. Digitalization is revolutionizing the maintenance and operation of industrial gas turbines. The use of sensors, data analytics, and machine learning allows for real-time monitoring and analysis of operational data.

3.4.1 Predictive Maintenance. In the realm of gas turbines, predictive maintenance has emerged as a cornerstone strategy aimed at enhancing operational efficiency and minimizing downtime. By harnessing advanced data analytics and machine learning techniques, it enables proactive identification of potential faults [144–146] and deterioration, thereby optimizing maintenance schedules and extending the lifespan of critical components. This approach not only improves reliability, but also reduces operational costs, making it indispensable in modern turbine management practices [52].

Machine learning algorithms are revolutionizing maintenance strategies by analyzing historical and real-time data to identify anomalies and recommend preventive actions, minimizing the risk of equipment failures [10]. Traditional maintenance approaches in the oil and gas industry often prove inefficient, as highlighted by Jambol et al. [147], who advocate for AI-driven predictive maintenance as a transformative solution. This approach leverages machine learning to anticipate maintenance needs, significantly reducing downtime, cutting costs, and improving equipment reliability. While the benefits of AI adoption are substantial, the necessary strategies and associated challenges also need to be examined.

To address the intricate problem of gas turbine fault diagnosis, a temporal convolutional network–autoencoder model has been developed. This innovative model combines the strengths of temporal convolutional networks and multi-head attention mechanisms to accurately detect anomalies in vibration data, offering a high level of precision [148].

To further enhance gas turbine performance and reliability, advanced models have been introduced for the online evaluation of damage to components subjected to complex cyclic loads. These models employ sophisticated cycle counting techniques and incorporate variations in temperature and stress, enabling more accurate assessments of fatigue damage [149,150]. Additionally, a probabilistic methodology has been applied to estimate the low-cycle fatigue life of a gas turbine compressor vane carrier under varying operating conditions. This approach integrates sensitivity analysis and surrogate modeling to refine life predictions and adapt maintenance planning [151].

3.4.2 Digital Twins and Remote Operation. Real-time data enable precise control and optimization of turbine operation, leading to improved efficiency and reliability. Digital twins, which are virtual representations of physical turbines, allow operators to simulate various scenarios and adjust operation accordingly [12,152].

Cruz-Manzo et al. developed a sensor fault diagnostic system specifically for industrial gas turbines. Integrated with a real-time digital twin, this system uses redundancy from a reference engine model to provide accurate estimates of engine variables, thereby enhancing turbine reliability and availability by proactively addressing sensor failures [153].

Apostolidis et al. explored data-driven strategies such as condition-based and predictive maintenance in aviation, aiming to reduce maintenance costs and environmental impact by diagnosing and forecasting aircraft system conditions. Using the generalized additive model, their research accurately predicts the evolution of exhaust gas temperature based on NASA's synthetic datasets. This highlights the interpretability of the model and underscores the importance of reliable data for certifiable AI applications in gas turbine diagnostics [154].

Furthermore, advances in remote monitoring technology now enable centralized control of industrial gas turbines, minimizing the need for on-site personnel and facilitating rapid responses to operational issues [150,155]. For the effect of start-up and shutdown cycles on transient temperature loading of rotor and stator of a small industrial gas turbine, a transient solver for single-shaft and twin-shaft gas turbines was applied [156]. Compressor and air-cooled turbine characteristics were computed with a through-flow solver and coupled by a simplified combustion model. The model includes gas control schemes for the transient operation of the gas turbine and a calibrated transient tip clearance model. It allows the creation of a database for various geometry modifications of the components to trace back deviations from expected engine behavior during real-time operation.

The German national project TurboGruen [157] introduced groundbreaking dynamic reduced order models (ROMs) for real-time prediction and management of critical industrial components.

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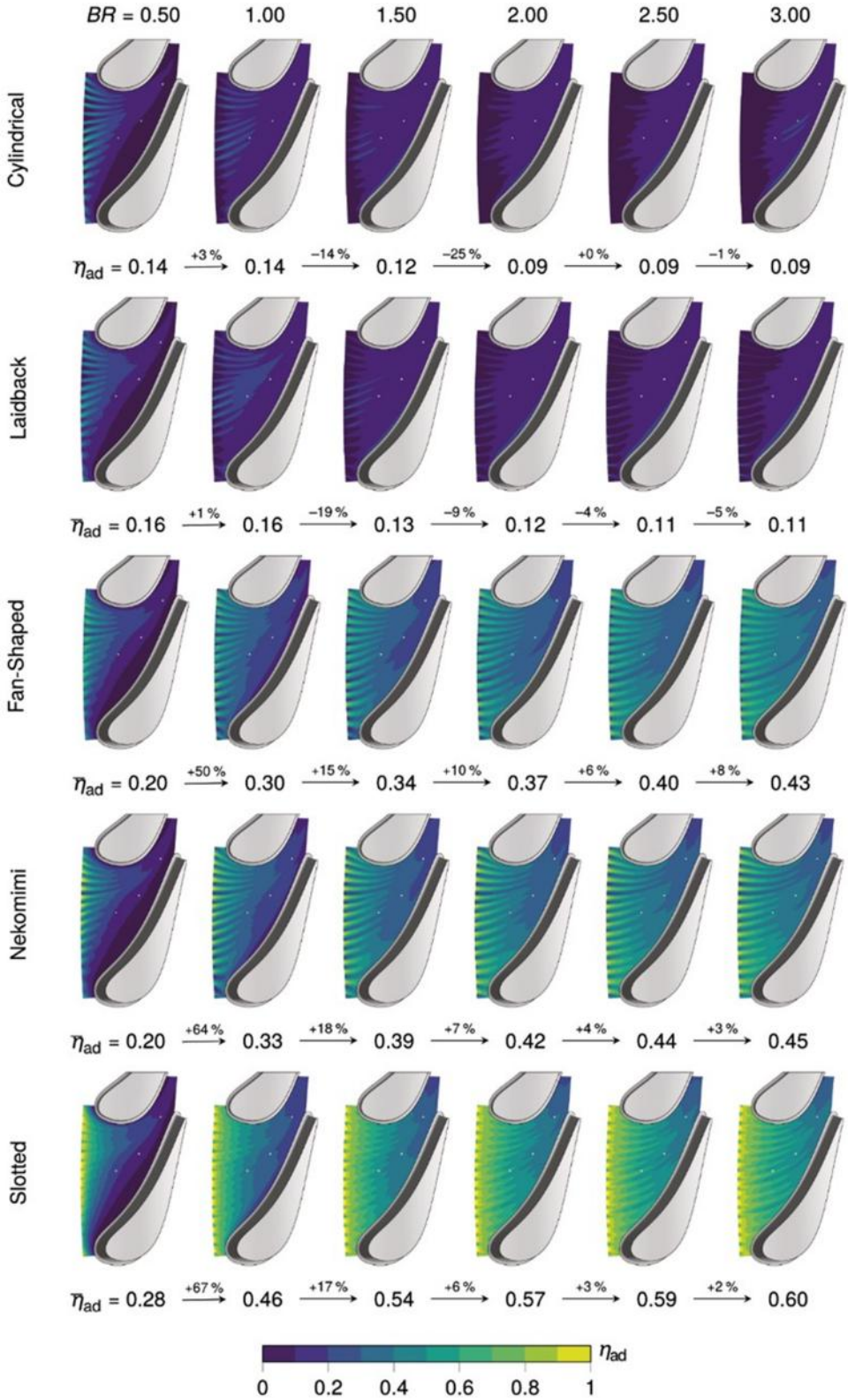

Fig. 16 Effect of slot design on cooling and flow [100]

It developed dynamic ROMs capable of predicting visco-plasticity in 3D fields, addressing stress, strain, and temperature under highly transient conditions. This advanced the understanding of material behavior under complex loads. By integrating these dynamic ROMs into a digital twin framework, precise predictions of damage accumulation and fatigue behavior were achieved, enhancing the reliability and lifespan of thermally highly loaded components (Fig. 17). Leveraging high-fidelity data, the ROMs delivered high accuracy while reducing simulation times by a factor of up to 100,000 compared to traditional finite element methods, enabling real-time monitoring and operational adjustments. A novel damage model based on Riedel's approach was

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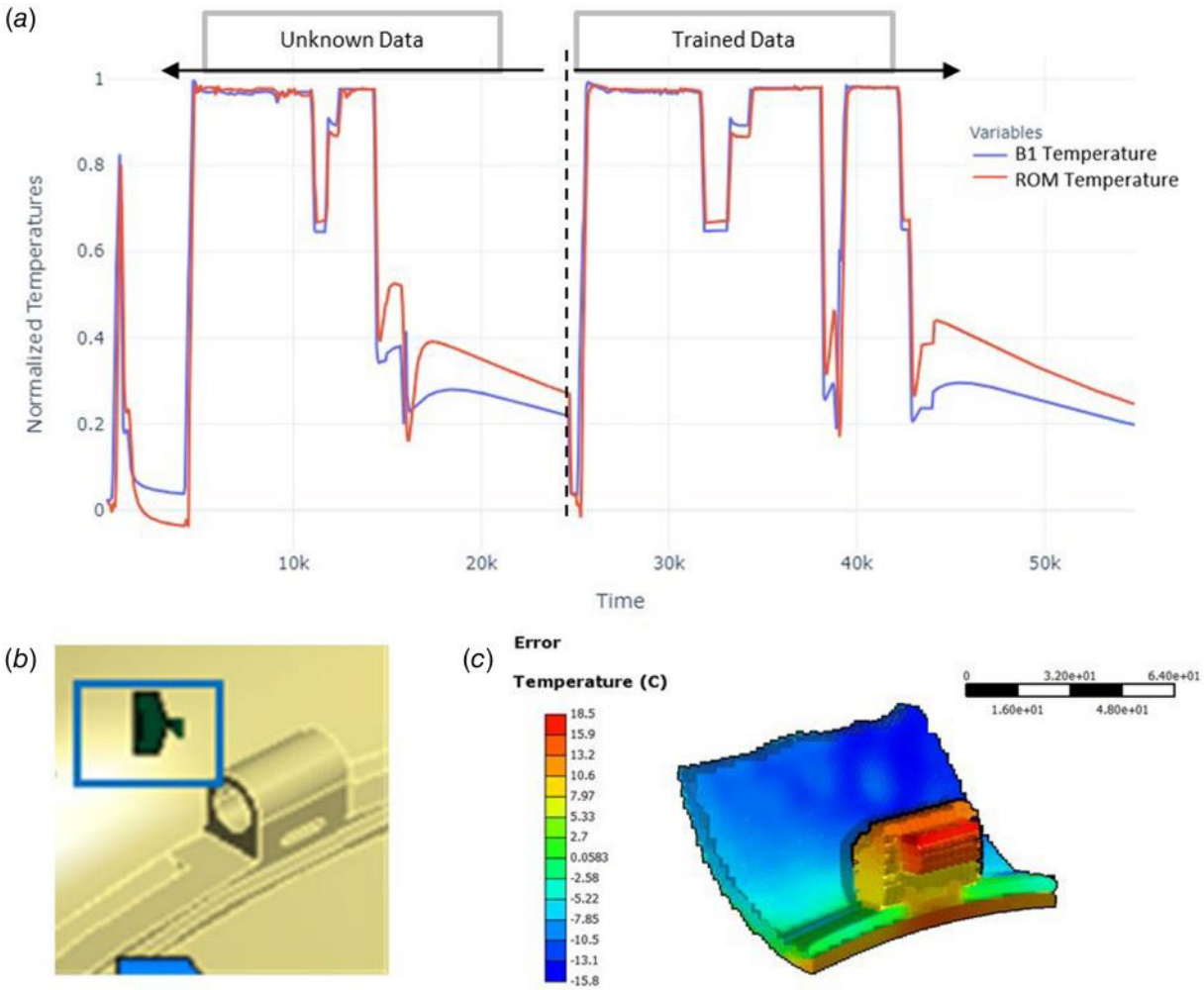

Fig. 17 ROM temperature predictions versus measurements: (*a*) normalized gas turbine signals over time (testing—2 days, resampled—1 s), (*b*) geometry, and (*c*) ROM error in steady-state operation

implemented. The novel approach enabled real-time performance assessments, better maintenance planning, and extended system lifecycles.

3.5 Integration of Renewable Energy. Industrial gas turbines play a crucial role in integrating renewable energy into the power grid. They provide the necessary flexibility to balance the intermittent power production from renewable sources.

3.5.1 Combined Heat and Power. Gas turbines of the considered power range are commonly used in CHP plants to generate both electricity and heat. By integrating renewable energy sources such as biogas [158], these systems can further enhance efficiency while reducing their environmental impact [18,19]. The European Union (EU) has outlined its strategy for advancing gas turbine and CHP technologies, with a focus on improving performance in the coming years [159]. Key goals of the early 2000s were achieving combined cycle efficiencies above 60%, industrial gas turbine system efficiencies of at least 50%, and small gas turbine efficiencies above 35%. Additionally, there was an emphasis on designing systems capable of using fuels with less than 25% of the heating value of natural gas. For CHP systems specifically, the primary objectives are to lower overall costs and develop biomass-fired systems with capacities greater than 40 kW [160].

Bespalov et al. have conducted research on assessing the life cycle of steam and gas turbine CHP plants, highlighting the potential for improving energy efficiency and minimizing environmental impacts [161]. Furthermore, variations in heat and power integration have been demonstrated between syngas-fueled and H_2-fueled CHP systems [162]. CH_4 steam reforming offers a method for recovering excess heat from steam in the form of syngas, although it reduces the electrical output of the CHP [163].

Bottoming organic Rankine cycles in CHP systems are particularly effective in capturing additional energy from low-temperature waste heat sources that would otherwise go unused, thus enhancing the overall efficiency and sustainability of the energy conversion [164,165] employing organic fluids like hydrofluorocarbons, fluorocarbons, and hydrocarbons as the working medium. Additionally, studies have examined the use of a transcritical carbon dioxide cycle to recover waste heat from gas turbines, with the goal of simultaneously producing electricity and cooling by utilizing carbon dioxide as the primary working fluid [166]. The feasibility of applying a pre-combustion carbon capture and storage (CCS) concept in CHP systems has also been investigated, offering another pathway toward reducing emissions and increasing system efficiency [167].

3.5.2 Hybrid Power Plants. Combining gas turbines with solar or wind power plants can enhance the stability and efficiency of the energy system. In hybrid power plants, gas turbines can ramp up or down quickly to offset the variability of renewable energy, ensuring a consistent power supply [168–170] (Fig. 18).

The performance of hybrid solar gas turbines is influenced by geographical and climate parameters, as these factors affect their

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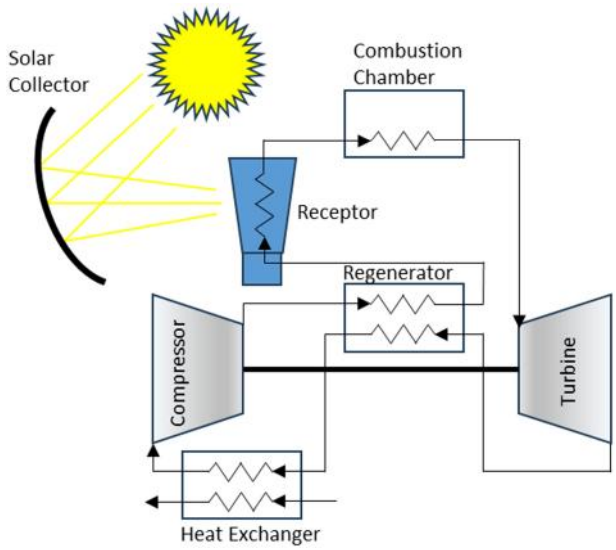

Fig. 18 Principle-hybrid Brayton cycle solar plant

overall efficiency and feasibility in different regions [171]. To enhance the design and application of such systems, advanced modeling approaches have been applied to micro gas turbines tailored for solar-hybrid operations, enabling more accurate predictions and optimizations [172]. Building on these concepts, a triple hybrid system has been developed to simultaneously generate electricity, heating, and cooling. This innovative system maximizes the use of exhaust gases from gas turbines, channeling them into absorption chillers for efficient cooling. At the core of this setup is a hybrid cycle that integrates a gas turbine with a fuel cell, serving as the primary actuator and driving both energy conversion and resource efficiency [173].

Based on an industrial gas turbine of the 12 MW class, a concept for a solar-hybrid combined cycle (SHCC) was published by Heide et al. [17]. For the study, a THM gas turbine with two combustor cans was modified to deliver the compressor exit flow to a heat exchanger where the air is heated by the high-temperature thermal energy from the solar collector before entering the combustion chambers (Fig. 19(*a*)). With a pressure ratio of 10 and moderate turbine inlet temperatures, the parameters are optimally suited for the SHCC cycle, and it can be operated without or only little extra-firing in a certain operating range (Fig. 19(*b*)). Plant concepts that could be adopted for large industrial gas turbines were proposed.

To address fluctuations in the grid, research has focused on enhancing the overall efficiency of power systems through the integration of gas turbine power plants and energy storage solutions, enabling a more consistent energy supply and grid stability [174]. This includes the development of a novel combustor concept designed specifically for hybrid electric gas turbines, which aims to improve the performance and emissions control [175]. Furthermore, Puppe et al. explored the incorporation of thermal energy storage into hybrid gas turbine systems, emphasizing the potential for reducing operating temperatures and improving efficiency during fluctuations in energy demand [16].

One notable development is the solid oxide fuel cell-gas turbine hybrid system, which combines electricity generation from solid oxide fuel cells with the utilization of waste heat and hot exhaust gases from gas turbines. This integration not only achieves exceptionally high overall efficiencies but also offers opportunities for producing both electricity and heat. Beyond the thermodynamic performance, economic operation modes for the system have been extensively analyzed to optimize feasibility and cost-effectiveness [176–178]. However, great care must be taken regarding fuel purity to avoid degradation of the fuel cell [179].

4 Global Perspectives and Market Trends

Having explored the technological innovations driving the evolution of industrial gas turbines, the focus now shifts to their role within the broader global energy landscape. This section examines market trends and regional perspectives to provide a comprehensive understanding of how these advances align with worldwide energy demands and policies.

4.1 Global Energy Market. The global energy market is undergoing significant changes. The transition to renewable energy and the increasing electrification of various sectors are influencing the demand for industrial gas turbines [22].

According to the International Energy Agency, global additions of renewable power capacity reached a record 507 GW in 2023, 50% more than in 2022. Solar energy accounted for three-quarters and wind energy for one-quarter of this increase [21]. This development naturally raises the question about the future market development of industrial gas turbines.

Forecasts indicate continued growth in renewable energy capacity. Consequently, gas turbine capacity will also keep expanding to support renewables, meet peak demand, and replace retiring coal plants. To reduce CO_2 emissions, many gas turbines can now operate with up to 50% H_2 (by volume). By 2030, advanced

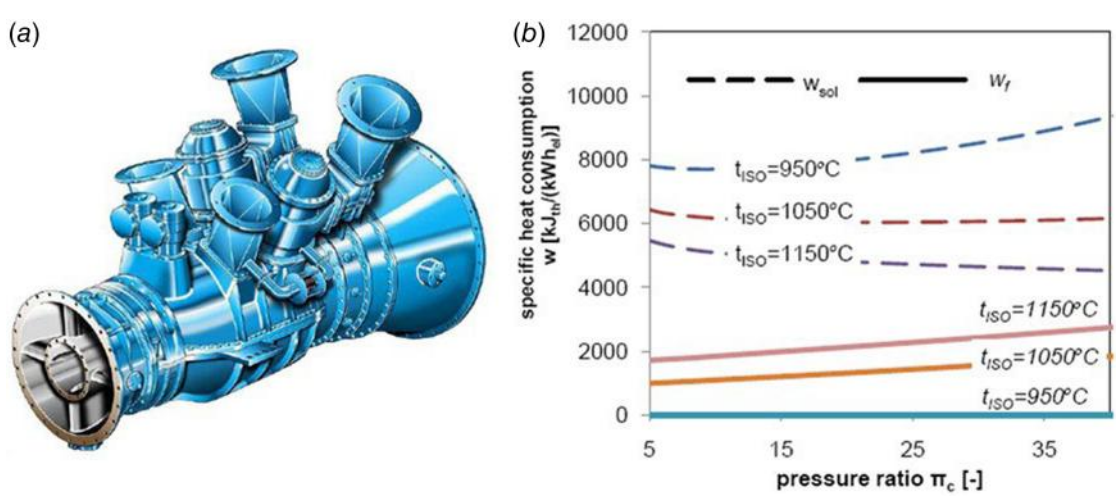

Fig. 19 THM gas turbine for external heating in an SHCC: (*a*) modification [17] and (*b*) specific heat consumption of an industrial gas turbine as prime mover in an SHCC [17]

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DLE combustion technology is expected to enable 100% H_2 use. Additionally, carbon capture and storage technologies are being developed for both retrofitting and new gas turbine power plants [21].

In 2023, overall gas turbine MW capacity orders decreased by 12.6%, reflecting the cyclical nature of the market. Despite this decline, the total number of units sold increased by 3%, with unit orders rising steadily from 389 units in 2019 to 453 units in 2023. This suggests a halt, if not a reversal, in the downward trend of gas turbine sales observed from 2010 to 2019, with a shift to smaller units. Gas turbines are expected to have long-term potential in the renewable offset and peaking market. From 2019 to 2023, global gas turbine sales fluctuated between 39,975 MW and 44,101 MW [21].

Over the next 10 years, USD 137 billion is projected to be spent on new turbines and turbine packages, excluding auxiliary systems and related equipment. With over 39,000 industrial gas turbines currently in operation globally, the overhaul and repair aftermarket is projected to be worth more than USD 261 billion. By the end of 2024, a 17% increase in unit orders in the Electric Power Utility sector and a 5% increase in unit orders in the O&G sector are anticipated [21].

The global gas turbine market was estimated in 2023 to be valued between USD 10.19 billion and USD 20.79 billion. Growth forecasts for the market vary slightly, with the compound annual growth rate ranging between 3.6% and 4.0%. By 2030 or 2032, the market is expected to grow to values between USD 22.5 billion and USD 29.36 billion [180–183].

While global market forecasts for gas turbines are positive, regional variations in policy frameworks significantly affect demand. For instance, the European Union supports gas turbine upgrades and hydrogen-readiness through subsidies and regulatory incentives, whereas in many Asia–Pacific countries the focus remains on cost-effective coal-to-gas switching rather than large-scale hydrogen adoption. In North America, policies such as the Inflation Reduction Act promote low-carbon fuels, yet infrastructure bottlenecks and uncertainty about carbon pricing slow deployment. These differences mean that the global growth projections may mask substantial regional disparities in investment timing and technology adoption.

The leading gas turbine OEMs for MWt orders in 2023 were Mitsubishi Power with 15,431 MW (35%), Siemens Energy with 10,689 MW (24%), and General Electric with 7015 MW (16%) (Fig. 20).

As mentioned earlier, aeroderivative gas turbines are aero-engines that are modified for industrial use, with about 80% interchangeable components between the engines and their derivatives. They offer quick start and shutdown capabilities and can cycle frequently without impacting maintenance schedules. From 2017 to 2023, they held a 4% market share (10,548 MW) in the engine power unit (EPU) sector and a 42% market share (19,436 MW) in the O&G sector. Over the next 5 years, aeroderivative turbine equipment sales are expected to reach USD 12.5 billion, with an annual maintenance market valued at USD 3.3 billion. These turbines are particularly in demand for peaking power and backup due to the increasing integration of renewables in the energy mix [21].

Heavy frame gas turbines are robust, industrially designed machines. They dominated the EPU sector with a 95% market share (217,014 MW) and held a 44% market share (20,106 MW) in the O&G sector from 2017 to 2023. Over the next 5 years, heavy frame turbine equipment sales are projected to reach a total of USD 48.6 billion, with an annual maintenance market of USD 19.6 billion.

Light industrial gas turbines combine features of both aeroderivatives and heavy frame turbines. They captured 14% of the market share and 38% of units (600 units) in the O&G sector from 2017 to 2023. Projected sales for the next 5 years are USD 4.4 billion, with an annual maintenance market of USD 652 million [21].

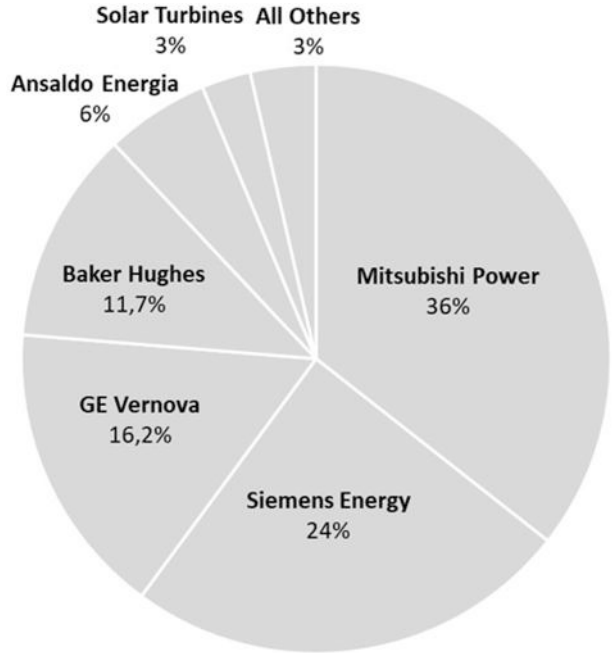

Fig. 20 2023 OEM MW market share [21]

4.2 Global Perspectives on Industrial Gas Turbines. As the share of energy produced from renewable resources grows, thermal power plants must increasingly shift from providing base-load power to offering fluctuating back-up power in order to stabilize and regulate the grid. Since they were not originally designed for this role, these frequent fluctuations may lead not only to increased wear and tear, but more importantly, to reduced efficiency and, consequently, higher greenhouse gas emissions per unit of electricity produced. To utilize existing transmission infrastructure, retrofitting existing plants and replacing coal-fired plants with newly built gas plants is considered an attractive option [184].

Significant load changes can be limited through load leveling, i.e., storing power during periods of low demand and delivering it during periods of high demand. Due to the increased integration of renewable power generation and its impact on the stability of the power grid, there is a growing need for long-term storage solutions. Battery technologies still face significant challenges, including dependence on rare earth elements and other raw materials sourced from countries with varying levels of political stability, questionable labor conditions, and a high carbon footprint. These factors could severely affect the overall CO_2 life cycle balance when batteries are employed as a storage technology. Therefore, chemical storage, i.e., using electrical energy to produce energy rich and stable chemical compounds, is widely seen as a promising path. Among those conversion routes, production of green H_2 via electrolysis of water is the most promising [185].

To address the challenges of adapting to the new energy conversion infrastructure, European gas turbine manufacturers have made a commitment to align with the trend of driving the transition to renewable-gas power generation and to promote the development of combustion technologies for green gases in their gas turbines [186] (Fig. 21). In their statement of 2020 they committed to:

(1) Delivering gas turbines in Europe that are capable of operating with **renewable gases** for power generation
(2) Providing customers with gas turbines capable of handling a **20% H_2 share (by volume)**
(3) Offering **retrofit** solutions for existing power plants to enable operation with renewable gases
(4) Meeting customer demand for gas turbines capable of running on **100% H_2** as part of a long-term investment strategy by 2030:"By providing the technologies that will drive

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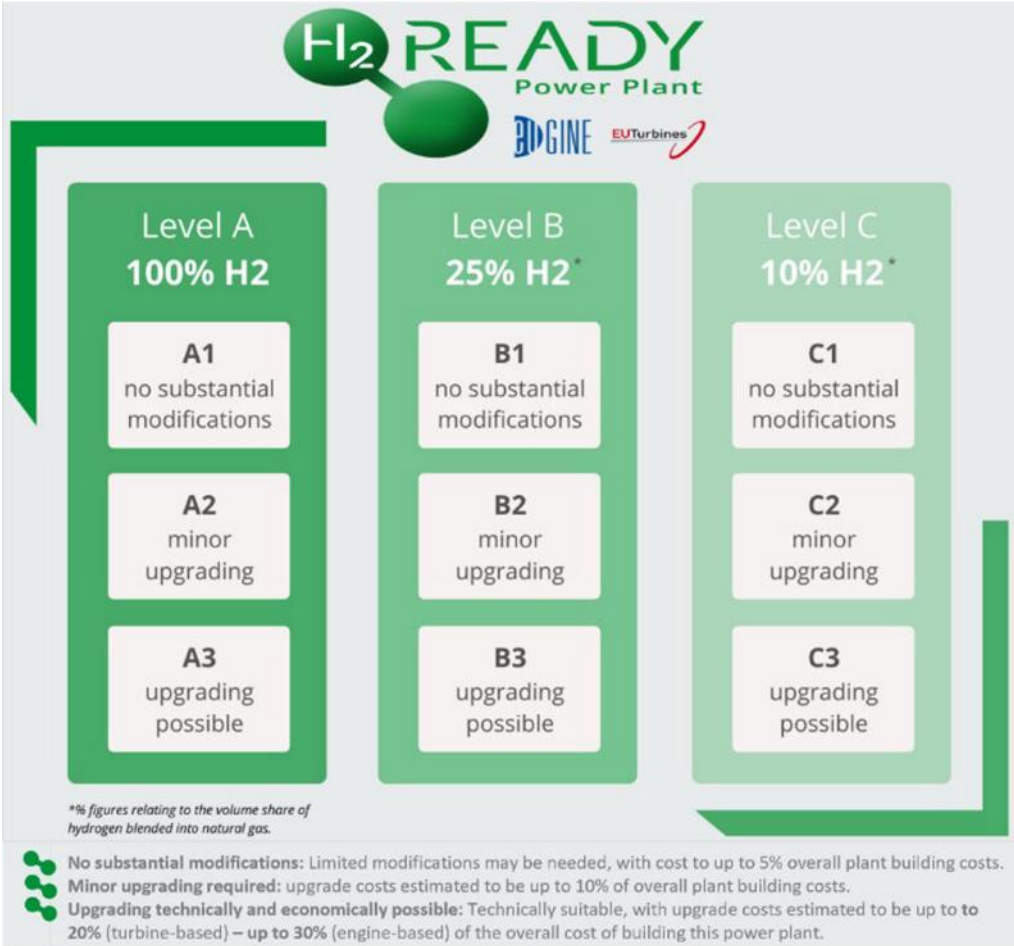

Fig. 21 Definitions of H_2 readiness by EUTurbines board members

the transition to fast-adapting, cost-efficient and renewable-gas-based power generation, we ensure that investments in gas turbines will neither become stranded assets nor contribute to a carbon lock-in."

A prerequisite for the decarbonized operation of H_2-ready gas turbines is the availability of large amounts of H_2 and prioritization of efforts to increase H_2 production from renewable sources, particularly through the establishment of large-scale power-to-gas plants. Current state of the art is production of H_2 out of fossil sources using steam CH_4 reforming to produce so-called blue H_2, if CCS is applied, at affordable prices [187].

With the aspired share of 85% renewable power by 2050, an abundance of renewable power generated from on- and offshore wind parks will be available for storage and used to produce H_2 at an industrial scale. H_2 transport lines need to be established beyond existing natural gas pipelines, and new compressor stations for H_2 must be developed because of the thermodynamic properties and high Mach numbers of H_2, which necessitate new designs [188]. While H_2 and renewable energy will play an increasingly significant role in the future energy mix, natural gas turbines continue to be crucial components of power generation, particularly in regions with an established gas infrastructure and growing electricity demand.

In North America, the industrial gas turbine market is driven by the need for reliable and flexible power generation. This region is known for its mature energy infrastructure and a strong emphasis on reducing carbon emissions. In 2023, the US added 9132 MW of new gas turbine capacity. This includes 7376 MW in combined cycle plants, 1756 MW in simple cycle systems, and 142 MW in gas-fired reciprocating engines [21]. Due to rising power demand, there have been delays in the planned closures of coal plants in states such as Kansas, Nebraska, and Oklahoma [23], which offers bright opportunities for replacement with gas-powered plants.

Natural gas remains the dominant fuel for power generation in the US, providing 43% of the energy in 2023, according to the IEA. Following natural gas, nuclear energy accounts for 18%, coal for 16%, wind for 10%, and solar for 4%. The high usage of natural gas is attributed to its abundant and low-cost supply. Combined cycle plants, which offer up to 64% efficiency, represent the largest segment for gas-fired plants, providing low-cost base and intermediate load power. Simple cycle gas turbines are the most flexible, catering to peaking and backup needs for renewable energy sources [21].

The electricity consumption in the Asia–Pacific region is expected to grow at a high rate, driven by industrialization, increasing energy demand, and the rising need for clean technologies. As the region heavily relies on coal, government initiatives to reduce emissions are leading to a greater adoption of gas turbines, particularly in Japan, China, Australia, South Korea, and India. China's 14th Five-Year Plan supports this transition with favorable policies for new gas power plant projects [189].

In the European Union, policies and national regulations provide strong support for renewable energy and low-emission technologies. Grants, subsidies, and research funding are available for projects that contribute to energy transition goals. Several projects focusing on gas turbine technologies are on the way or planned, e.g., the EU-funded projects HYFLEXPOWER [127] and Flex4H2 [128], and for the 1.4 GW Magnum power plant (30% by volume co-firing) in the Netherlands [190].

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Worldwide, around 3500 MW of H_2-fired power plant capacity is estimated to exist by 2030 [182]. Around 85% of these projects focus on the use of H_2 in combined-cycle or open-cycle gas turbines. In the initial phase of most projects, a co-firing share in the range of 5–10% in energy terms (15–30% volumetric) shall be achieved as a prerequisite for the move to higher shares and in some cases even 100% H2 firing in the longer term. These projects are mainly located in the Asia–Pacific region (40%), Europe (33%), and North America [20].

Despite the growing number of hydrogen projects, the economic viability of hydrogen-fired power remains uncertain. The levelized cost of green hydrogen remains two to five times higher than natural gas on an energy-equivalent basis, even with carbon pricing [191]. Large-scale electrolyzer projects are capital-intensive, and their competitiveness depends on further cost reductions in renewable electricity and advances in electrolysis efficiency. Without sustained policy support, risk-sharing mechanisms, and infrastructure for H_2 transport and storage, widespread hydrogen adoption in gas turbines may lag behind expectations, potentially leading to stranded assets or higher system costs.

Currently, H_2 use for electricity generation amounts less than 0.2% in the worldwide electricity generation mix, and H_2 and NH_3 combined may be also limited to 2% of global electricity generation by 2050 [192]. Whether H_2 will become an important backbone of flexible power generation to bridge longer periods of dark doldrums will heavily depend on the availability and price of H_2 production utilizing excess electricity generated by renewable sources. Looking at the EU, even with ambitious ramp-up scenarios, the anticipated need for residual electrical power can only be met to a moderate extend (2050: about 100 GW of 345 GW) due to high production cost and low round-trip efficiency of less than 35% with gas turbine-driven plants [118]. The situation is similarly challenging in the US [183].

The growing energy demand of data centers creates a new market for industrial gas turbines. Their need for very high reliability (Tier IV, >99.995% availability) [193] and cleaner, longer-duration back-up power is driving a shift away from diesel generators. Gas turbines offer high power density, fast start capability, and the option to run on natural gas or hydrogen blends, enabling lower emissions and future decarbonization. This segment is expected to grow particularly in North America, Northern Europe, and Asia–Pacific, supported by partnerships between OEMs [194,195], utilities, and data center operators [196–198].

5 Outlook and Technology Perspectives

Concluding the analysis of global perspectives and market trends, this section explores the future prospects and challenges for industrial gas turbines. It addresses emerging opportunities and key obstacles in adapting to evolving energy demands and advancing technologies.

More than a decade ago, the development of H_2-fired gas turbines was in focus in connection with CCS technologies. Oxy-fuel combustion was considered one of the most promising options [199]. With fading focus on CCS in power generation plants, the development of H_2-fired gas turbines has slowed down in favor of increase of performance and operational flexibility as a response to a growing fleet of renewables. Further increase of volatile renewables as a countermeasure to greenhouse gas emissions causes a growing demand for large-scale energy storage. Among these storage options, power-to-gas is best suited for long-term storage of large capacities. The direct combustion of H_2, its carriers, and H_2-enriched gases has garnered increasing attention, with gas turbine technologies evolving from their former role as bridging technologies to becoming a major driver of the transition to a renewable energy system.

To avoid that, new installations will end up as stranded assets in a future energy system dominated by renewables, and all OEMs are working on solutions to fully replace natural gas with H_2 or its derivatives. While much of the development effort is naturally focused on DLE combustion systems, the fuel switch necessitates modifications across all core engine components.

As pointed out in the preceding section, small- to medium-sized gas turbines have already been developed for gas mixtures with up to 100% H_2 for special applications. In some regions, the OEMs' roadmaps are adjusted to incentives provided by strategies on a national level. One example is Japan, where a H_2 strategy has been set up as early as 2017 covering all energy-related sectors, including automobility [200]. It states:

> "Electricity can be generated through combustion, both as pure hydrogen and as a mixture with natural gas or coal. […] Power generation from hydrogen combustion is expected to be more cost effective than electricity generation by stationary fuel cells due to scalability. […] Conversely this means that power generation becomes less advantageous in small-scale distributed power generators. To benefit from operation as a large-scale plant, it must also secure sufficient volumes of hydrogen fuel."

In response to the national strategy, Japanese gas turbine manufacturers have developed combustion systems capable of handling increased H_2 content. It has become evident that large volumes of H_2 will need to be imported, and the world's first liquefied H_2 carrier completed its demonstration voyage in 2022 [201]. The first commercially operated gas turbine with a DLN-combustion system capable for 100% H_2 by **KHI** was installed in Kobe, Japan, (M1A-17D Series, ~1 MW_{el}), including a receiver station for blue H_2 imported from Australia [202]. This concept is based on so-called Micro-Mix diffusion flames and complies with statutory NO_x emission regulations [203]. **Mitsubishi Power Systems (MPS)** announced that their combustion system of a large-scale heavy-duty gas turbine successfully passed a firing test using a mixture of LNG and 30% H_2 (by volume) [204]. In 2021, **Mitsubishi Heavy Industries** started the development to enable a 40 MW-class gas turbine (H-25 Series) for the combustion of 100% NH_3-based fuel. Currently, feasibility studies are being conducted in cooperation with Thailand's major utility company [205]. **GE** has developed a new combustor, designated DLN 2.6e, for its 9HA gas turbines, which is reportedly capable of operating on a natural gas blend containing approximately 50% H_2 (by volume) [206]. For the GT26/36-series development activities are ongoing at **Ansaldo** aiming at even higher H_2 fuel shares (HORIZON EUROPE-Project Flex4H2 [128]). **Baker Hughes (BH)** reported on the delivery of a NovaLT16 gas turbine for a green H_2 power plant facility in South Australia claiming combustion of 100% H_2 [206]. **Siemens Energy** is upgrading the combustion systems across its gas turbine fleet and has recently published details on the SGT-400 industrial gas turbine [206] as well as the HORIZON 2020 project HYFLEXPOWER [127].

In the future, gas turbine-powered CHP plants are expected to exhibit promising development and market potential. In the near to mid-term, they are likely to replace coal-fired power plants and boilers on a large scale to reduce greenhouse gas emissions associated with natural gas, thereby supporting the transitional phase toward zero-carbon power generation.

For alternative fuels with significantly different heating values, adjustments in the matching between turbine and compressor are required. To maintain fuel flexibility, the compressor's operating range may need to be expanded, making an increased surge margin a key focus of current research. In this context, design features from aero-engine compressors—such as casing treatments—may also be increasingly adopted in stationary industrial gas turbine compressors.

Apart from aerodynamic considerations in the expansion turbine, the exhaust gas composition changes and contains more H_2O instead of CO_2. As a consequence, the cooling flow demand will rise. The aero-thermal and mechanical layout of the entire turbine needs to be revised.

The industry will adapt to the demands of a highly volatile power generation market, requiring gas turbines to be flexible both in

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terms of fuel and operation. They will primarily run when demand exceeds renewable generation. Under these circumstances, customers will prioritize reliable and robust engines at a high level of availability.

Though the production cost of green H_2 is expected to decrease in the future, the low round-trip efficiency of the H_2 re-electrification chain—including production, compression, and distribution—will continually demand high cycle and component full and part load efficiencies while ensuring operational flexibility as a key driver of future research to allow a large number of start and shut-down cycles. This means that all disciplines—including aerodynamics, aero-thermal, and mechanical integrity, as well as the development of new materials—will continue to play a crucial role in the advancement of industrial gas turbines.

6 Conclusion

The aim of this article is to provide a perspective on the future role of industrial gas turbines in global energy trends. The efficiency, fuel flexibility, and stability of industrial gas turbines make them indispensable for power generation and industrial applications. Advances in materials, aerodynamics, and combustion technologies have significantly improved their performance, while digitalization and predictive maintenance enhance operational efficiency. The shift toward alternative fuels and integration with renewable energy sources further solidify their role in a sustainable energy future.

In conclusion, industrial gas turbines will continue to be essential in the evolving energy landscape. Advancements in efficiency, sustainability, and digitalization will ensure their relevance in a decarbonized future. Their integration with renewable energy and improved fuel flexibility will define their role in providing reliable and adaptable power solutions.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

No data, models, or code were generated or used for this article.

References

[1] Saravanamuttoo, H. I. H., Rogers, G., and Straznicky, P. V., 2009, *Gas Turbine Theory*, 6th ed., Pearson Prentice Hall, Harlow, Munich.
[2] Ricci, M., Benvenuto, M., Mosele, S. G., Pacciani, R., and Marconcini, M., 2021, "Impact of Compressor Flexibility Improvements on Heavy-Duty Gas Turbine for Minimum and Base Load Conditions," SSRN. .
[3] Blakey-Milner, B., Gradl, P., Snedden, G., Brooks, M., Pitot, J., Lopez, E., Leary, M., et al., 2021, "Metal Additive Manufacturing in Aerospace: A Review," Mater. Des., **209**, p. 110008.
[4] Risse, J., 2019, *Additive Fertigung der Nickelbasis-Superlegierung IN738LC Mittels Selektivem Laserstrahlschmelzen*, RWTH Aachen University, Aachen, Germany.
[5] Faqih, M., Omar, M. B., Ibrahim, R., and Omar, B. A. A., 2022, "Dry-Low Emission Gas Turbine Technology: Recent Trends and Challenges," Appl. Sci., **12**(21), p. 10922.
[6] Marwaha, G., and Kohn, J., 2019, "Predictive Maintenance of Gas Turbine Air Inlet Systems for Enhanced Profitability as a Function of Environmental Conditions," Abu Dhabi International Petroleum Exhibition & Conference (SPE-197814-MS).
[7] Liu, Y., and Jiang, X., 2022, "Towards Predictive Maintenance of a Heavy-Duty Gas Turbine a New Hybrid Intelligent Methodology for Performance Simulation," Ann. Conf. PHM Soc., **14**(1), pp. 1–9.
[8] Brahimi, L., Hadroug, N., Iratni, A., Hafaifa, A., and Colak, I., 2024, "Advancing Predictive Maintenance for Gas Turbines: An Intelligent Monitoring Approach With ANFIS, LSTM, and Reliability Analysis," Comput. Ind. Eng., **191**, p. 110094.
[9] Berghout, T., Amirat, Y., Bounceur, A., and Benbouzid, M., 2024, "AI-Based Gas Turbine Multi-component Health Prognosis Via Recurrent Expansion of Gas Path Parameters," ICCAD 24—The 8th International Conference on Control, Automation, Paris, France, May 15–17.
[10] Bangert, P., 2020, "Predictive Maintenance for Gas Turbines," International Petroleum Technology Conference (D023S039R003).
[11] Abdelfettah, M., and Oualid, B., 2024, "Predicting the Remaining Useful Life of a Gas Turbine Based on an Exponential Degradation Model," J Fail. Anal. Preven., **24**(3), pp. 1295–1309.
[12] Schulze, R., and Knödl, S., 2023, Digital Twin Technology for Compressor and Turbine Efficiency Evaluation—Application to Energy and GHG Savings."
[13] Panov, V., and Cruz-Manzo, S., 2020, "Gas Turbine Performance Digital Twin for Real-Time Embedded Systems," Volume 5: Controls, Diagnostics, and Instrumentation; Cycle Innovations; Cycle Innovations: Energy Storage, Turbo Expo: Power for Land, Sea, and Air, Virtual, Sept. 21–25.
[14] Panov, V., and Cruz, S., 2021, "Embedded, Edge and Cloud Computing for Gas Turbine Digital Twins."
[15] Lim, J., Perullo, C. A., Milton, J., Whitacre, R., Jackson, C., Griffin, C., Noble, D., et al., 2021, "The EPRI Gas Turbine Digital Twin—A Platform for Operator Focused Integrated Diagnostics and Performance Forecasting," Volume 4: Controls, Diagnostics, and Instrumentation; Cycle Innovations; Cycle Innovations: Energy Storage; Education; Electric Power, Turbo Expo: Power for Land, Sea, and Air, Virtual, Online, June 7–11.
[16] Puppe, M., Giuliano, S., Buck, R., Krüger, M., Lammel, O., Saidi, K., Felsmann, C., et al., 2015, "Hybrid High Solar Share Gas Turbine Systems With Innovative Gas Turbine Cycles," Energy Proc., **69**, pp. 1393–1403.
[17] Heide, S., Gampe, U., Orth, U., Beukenberg, M., Gericke, B., Freimark, M., Langnickel, U., et al., 2010, "Design and Operational Aspects of Gas and Steam Turbines for the Novel Solar Hybrid Combined Cycle SHCC®," Proceedings of the ASME Turbo Expo 2010, ASME Turbo Expo 2010: Power for Land, Sea, and Air, ASME, Glasgow, UK, June 14–18, pp. 465–474.
[18] Vogel, T., Oeljeklaus, G., Görner, K., Polklas, T., and Frekers, C., 2016, Comparative Study of Gas Engines and Gas Turbines in Cogeneration (CHP), Using the Example of a Typical Public District Heat Distribution Network.
[19] Mo, D., Liu, Y., Roumeliotis, I., Li, Y., and Nalianda, D., 2023, "Review of Gas Turbine Combined Heat and Power Systems," The Proceedings of the 2021 Asia-Pacific International Symposium on Aerospace Technology (APISAT 2021), Jeju, South Korea, November 15-17, 2021, Springer, Vol. 913, pp. 1279–1298.
[20] Wiedermann, A., Jansohn, P., Payrhuber, K., Bernstrauch, O., Cammarata, A., Coda Zabetta, E., and Georgiev, M., 2024, *Ramp-Up and Role of Hydrogen-Based Power Generation—WG3 Focus Paper—ETIP SNET*, European Commission; Directorate-General for Energy, Publications Office of the European Union, European Union.
[21] Gas Turbine World. "Gas Turbine Market Forecast." https://gasturbineworld.com/market-forecast/, Accessed July 11, 2024.
[22] Tracey, M., and Robb, A., 2023, "Turbomachinery International Handbook," *Glob. J. Energy Equip.*, Vol. 63, Turbomachinery International Magazine, Houston, TX.
[23] Turbomachinery Magazine. "Gas Turbine Sales Report: 2023 Holds Its Own." https://www.turbomachinerymag.com/view/gas-turbine-sales-report-2023-holds-its-own, Accessed July 17, 2024.
[24] ASME, 2007, "The World's First Industrial Gas Turbine Set—GT Neuchatel.".
[25] MIT. "Early Gas Turbine History." http://web.mit.edu/aeroastro/labs/gtl/early_GT_history.html#1939, Accessed June 13, 2024.
[26] NASA. "Gas Turbine Propulsion." https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/turbine-engines/#gas-turbine-jet-engine, Accessed June 13, 2024.
[27] Bakken, L. E., Jordal, K., Syverud, E., and Veer, T., 2004, "Centenary of the First Gas Turbine to Give Net Power Output: A Tribute to Ægidius Elling," Volume 2: Turbo Expo 2004, Turbo Expo: Power for Land, Sea, and Air, Vienna, Austria, June 14–17, pp. 83–88.
[28] Stodola, A., 1924, *Dampf- und Gasturbinen*, 6th ed., Springer, Berlin, Germany.
[29] Boyne, W. J., ed., 1979, *The Jet Age*, Smithsonian Institution Press, Washington, DC.
[30] Dixon, S. L., and Hall, C. A., eds., 2010, *Fluid Mechanics and Thermodynamics of Turbomachinery*, Elsevier, New York.
[31] Driscoll, M. J., McFetridge, E. M., Patterson, J. S., and See, C. A., 2011, "Removals for Cause: A 35-Year Assessment of LM2500 Engine Removals by the United States Navy," Volume 4: Cycle Innovations; Fans and Blowers; Industrial and Cogeneration; Manufacturing Materials and Metallurgy; Marine; Oil and Gas Applications, ASME 2011 Turbo Expo:

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Turbine Technical Conference and Exposition, ASME DC, Vancouver, BC, Canada, June 6–10, pp. 799–808.
[32] Gunston, B., 2010, *The Development of Jet and Turbine Aero Engines*, 4th ed., Stephens, Sparkford, UK.
[33] Eckardt, D., 2014, *The Development of the Power Generation Gas Turbine at BBC—ABB—Alstom*, 2nd ed. ed., De Gruyter Oldenbourg, München, Germany.
[34] Saidi, K., 2021, "Industrial CHP Applications—Gas Turbines & Engines." https://app.gotowebinar.com/unified/index.html#/webinar/7431801404153780 05/attend/3387861851642901597, Accessed January 15, 2025.
[35] Vilag, V., Vilag, J., Carlanescu, R., Mangra, A., and Florean, F., 2019, "CFD Application for Gas Turbine Combustion Simulations," *Computational Fluid Dynamics Simulations*, IntechOpen, Rijeka, Croatia. Ch. 11.
[36] Ji, G., and Zhu, J., eds., 2019, *Computational Fluid Dynamics Simulations*, IntechOpen, Rijeka, Croatia.
[37] Zhou, H., Xue, J., Gao, H., and Ma, N., 2024, "Hydrogen-Fueled Gas Turbines in Future Energy System," Int. J. Hydrogen Energy, **64**, pp. 569–582.
[38] Öberg, S., Odenberger, M., and Johnsson, F., 2022, "Exploring the Competitiveness of Hydrogen-Fueled Gas Turbines in Future Energy Systems," Int. J. Hydrogen Energy, **47**(1), pp. 624–644.
[39] Chiesa, P., Lozza, G., and Mazzocchi, L., 2005, "Using Hydrogen as Gas Turbine Fuel," ASME J. Eng. Gas Turbines Power, **127**(1), pp. 73–80.
[40] Linkenheil, C. P., Küchle, I., Kurth, T., and Huneke, F., 2017, "Flexibility Needs and Options for Europe's Future Electricity System," EUGINE—The European Engine Power Plants Association. Energy Brainpool. https://www.energybrainpool.com/sites/default/files/2022-12/flexibility_options-energy_brainpool-eugine-01_2017.pdf, Accessed October 25, 2024.
[41] Solar Turbines, 2024, "Products." https://www.solarturbines.com/en_US/products.html. Accessed October 25, 2024.
[42] Siemens Energy, 2024, "Reliable Gas Turbines." https://www.siemens-energy.com/us/en/home/products-services/product-offerings/gas-turbines.html. Accessed October 25, 2024.
[43] Wiedermann, A., "Aeroderivate," *Stationäre Gasturbinen*, 3rd ed., J. Seume, ed., Springer, Berlin, Heidelberg. (in German; Lechner, Ch.)
[44] Gas Turbine World. "Gas Turbine Costs $/kW." https://gasturbineworld.com/gas-turbine-costs-kw/, Accessed September 23, 2025.
[45] Teraji, D., 2005, "Industrial Application of Gas Turbines Committee," Industrial Application of Gas Turbines Committee, 16th Symposium on Industrial Application of Gas Turbines (IAGT), Banff, AB, Canada, Oct. 12–14.
[46] Mehervan, P. B., Axial Compressors," *Gas Turbine Engineering Handbook*, 2nd ed., Butterworth-Heinemann, Boston, Oxford, Auckland, Johannesburg, Melbourne, New Delhi.
[47] Wiedermann, A., Frank, D., Orth, U., and Beukenberg, M., 2011, "Computational and Experimental Analysis of an Industrial Gas Turbine Compressor," ASME 2011 Turbo Expo: Volume 7: Turbomachinery, Parts A, B, and C, ASME 2011 Turbo Expo: Turbine Technical Conference and Exposition, American Society of Mechanical Engineers (ASME), Vancouver, BC, Canada, June 6–10, pp. 319–329.
[48] Yoon, S., and Cargill, P., 2023, "Casing Treatment: Its Potential and Limitations," ASME J. Turbomach., **145**(4), p. 041011.
[49] Goinis, G., Voss, C., and Nicke, E., 2021, "The Potential of Casing Treatments for Transonic Compressors: Evaluation Based on Axial-Slot and Rotor Blade Optimization," 32nd Congress of the International Council of the Aeronautical Sciences (ICAS), Shanghai, China, Sept. 6–10.
[50] Kurz, R., Mendoza, R., Burnes, D., Saxena, P., and Alexander, S., 2018, "On Gas Turbine Safety in Offshore Operations," Volume 9: Oil and Gas Applications; Supercritical CO2 Power Cycles; Wind Energy, ASME Turbo Expo 2018: Turbomachinery Technical Conference and Exposition, American Society of Mechanical Engineers, Oslo, Norway, June 11–15.
[51] Aschenbruck, E., Cagna, M., Mueller, R., Orth, U., Spiegel, A., Wiedermann, A., and Wiers, S.-H., 2015, "Operating Experience With MAN's MGT6200 Gas Turbine," Proceedings of International Gas Turbine Congress 2015 Tokyo, Tokyo, Japan. (Paper 005).
[52] Mevissen, F., and Meo, M., 2019, "A Review of NDT/Structural Health Monitoring Techniques for Hot Gas Components in Gas Turbines," Sensors, **19**(3), p. 711.
[53] Kurz, R., and Brun, K., 2012, "Fouling Mechanisms in Axial Compressors," ASME J. Eng. Gas Turbines Power, **134**(3), p. 032401.
[54] Brun, K., Grimley, T. A., Foiles, W. C., and Kurz, R., 2015, "Experimental Evaluation of the Effectiveness of Online Water-Washing in Gas Turbine Compressors," ASME J. Eng. Gas Turbines Power, **137**(4), p. 042605.
[55] Assady, M., and Sammak, M., 2007, "Anti-Icing in Gas Turbines: Icing Conditions and Ice Formation Mechanisms," Proceedings of the International Gas Turbine Congress 2007, Tokyo, Japan. (Paper TS-068).
[56] Alsuwian, T., Amin, A. A., Iqbal, M. S., and Maqsood, M. T., 2023, "A Review of Anti-Surge Control Systems of Compressors and Advanced Fault-Tolerant Control Techniques for Integration Perspective," Heliyon, **9**(9), p. e19557.
[57] Blanchini, F., and Giannattasio, P., 2002, "Adaptive Control of Compressor Surge Instability," Automatica, **38**(8), pp. 1373–1380.
[58] Reed, R. C., 2006, "The Superalloys," *Fundamentals and Applications*, Cambridge University Press, Cambridge, New York, Melbourne, Madrid, Cape Town, Singapore, São Paulo.
[59] Bohn, D., Götte, J., Kusterer, K., and Liu, Y., 2024, "Thermodynamic Potential of Aluminum Oxide Ceramic Matrix Composite Application in the 1st Stage Vanes of Industrial Gas Turbines," Volume 13: Heat Transfer: General Interest/Additive Manufacturing Impacts on Heat Transfer; Wind Energy, ASME Turbo Expo 2024: Turbomachinery Technical Conference and Exposition, American Society of Mechanical Engineers, London, UK, June 24–28.
[60] Trosch, T., Strößner, J., Völkl, R., and Glatzel, U., 2016, "Microstructure and Mechanical Properties of Selective Laser Melted Inconel 718 Compared to Forging and Casting," Mater. Lett., **164**, pp. 428–431.
[61] Mevissen, F., and Meo, M., 2022, "Creep Detection of Hastelloy X Material for Gas Turbine Components With Nonlinear Ultrasonic Frequency Modulation," Mater. Charact., **191**, p. 112099.
[62] Mostafa, A., Picazo Rubio, I., Brailovski, V., Jahazi, M., and Medraj, M., 2017, "Structure, Texture and Phases in 3D Printed IN718 Alloy Subjected to Homogenization and HIP Treatments," Metals, **7**(6), p. 196.
[63] Waugh, I., Moore, E., Greig, A., Macfarlane, J., and Dick-Cleland, W., 2021, "Additive Manufacture of Rocket Engine Combustion Chambers Using the ABD®-900AM Nickel Superalloy."
[64] Bridges, A., Shingledecker, J., Clark, J., and Crudden, D., 2023, "Creep Analysis and Microstructural Evaluation of a Novel Additively Manufactured Nickel-Base Superalloy (ABD®-900AM)," ASME J. Eng. Gas Turbines Power, **145**(6), p. 061005.
[65] Torkaman, A., Vogel, G., and Houck, L., 2021, "Design, Development and Validation of Additively Manufactured First Stage Turbine Vane for F Class Industrial Gas Turbine," Volume 9A: Structures and Dynamics—Aerodynamics Excitation and Damping; Bearing and Seal Dynamics; Emerging Methods in Design and Engineering, Turbo Expo, Virtual, Online, June 7–11.
[66] Hossain, M. A., Ameri, A., Gregory, J. W., and Bons, J. P., 2021, "Experimental Investigation of Innovative Cooling Schemes on an Additively Manufactured Engine Scale Turbine Nozzle Guide Vane," ASME J. Turbomach., **143**(5), p. 051004.
[67] Takeishi, K., and Krewinkel, R., 2023, "Advanced Gas Turbine Cooling for the Carbon-Neutral Era," IJTPP, **8**(3), p. 19.
[68] Poprawe, R., Kelbassa, I., Walther, K., Witty, M., Bohn, D., and Krewinkel, R., 2008, "Optimising and Manufacturing a Laser-Drilled Cooling Hole Geometry for Effusion-Cooled Multi-layer Plates," Proceedings of the 12th ISROMAC, Honolulu, HI, Feb. 17–22, pp. 1–10.
[69] Acharya, S., "Gas Turbine Handbook," *Turbine Blade Aerodynamics*, U.S. Department of Energy. http://www.netl.doe.gov/research/coal/energy-systems/turbines/publications/handbook, Accessed December 9, 2024. (Chapter 4.3).
[70] Sieverding, C. H., Stanislas, M., and Snoeck, J., 1980, "The Base Pressure Problem in Transonic Turbine Cascades," ASME J. Eng. Gas Turbines Power, **102**(3), pp. 711–718.
[71] Langston, L. S., 2001, "Secondary Flows in Axial Turbines–A Review," Ann. N. Y. Acad. Sci., **934**(1), pp. 11–26.
[72] Sieverding, C. H., 1985, "Recent Progress in the Understanding of Basic Aspects of Secondary Flows in Turbine Blade Passages," ASME J. Eng. Gas Turbines Power, **107**(2), pp. 248–257.
[73] Rose, M. G., 1994, "Non-Axisymmetric Endwall Profiling in the HP NGV's of an Axial Flow Gas Turbine," Volume 1: Turbomachinery, ASME 1994 International Gas Turbine and Aeroengine Congress and Exposition, American Society of Mechanical Engineers, The Hague, The Netherlands, June 13–16.
[74] Dossena, V., Perdichizzi, A., and Savini, M., 1999, "The Influence of Endwall Contouring on the Performance of a Turbine Nozzle Guide Vane," ASME J. Turbomach., **121**(2), pp. 200–208.
[75] Thrift, A. A., Thole, K. A., and Hada, S., 2011, "Effects of an Axisymmetric Contoured Endwall on a Nozzle Guide Vane: Adiabatic Effectiveness Measurements," ASME J. Turbomach, **133**(4), p. 041007.
[76] Corriveau, D., and Sjolander, S. A., 2007, "Influence of Loading Distribution on the Off-Design Performance of High-Pressure Turbine Blades," ASME J. Turbomach., **129**(3), pp. 563–571.
[77] Pullan, G., Denton, J., and Curtis, E., 2006, "Improving the Performance of a Turbine With Low Aspect Ratio Stators by Aft-Loading," ASME J. Turbomach., **128**(3), pp. 492–499.
[78] Ames, F. E., Argenziano, M., and Wang, C., 2004, "Measurement and Prediction of Heat Transfer Distributions on an Aft-Loaded Vane Subjected to the Influence of Catalytic and Dry Low NOx Combustor Turbulence," ASME J. Turbomach., **126**(1), pp. 139–149.
[79] Denton, J. D., 1975, "A Time Marching Method for Two- and Three-Dimensional Blade to Blade Flows," Her Majesty's Stationary Office, ARC R&M 3775.
[80] Miki, K., and Liou, M.-S., 2017, "Computational Study of Combustor-Turbine Interactions," NASA TR 7987. https://ntrs.nasa.gov/api/citations/20170010713/downloads/20170010713.pdf, Accessed December 9, 2024.
[81] Lo Presti, F., Sembritzky, M., Winhart, B., Post, P., Di Mare, F., Wiedermann, A., Greving, J., et al., 2022, "Numerical Investigation of Unsteady Combustor Turbine Interaction for Flexible Power Generation," ASME J. Turbomach., **144**(2), p. 021003.
[82] Kluß, D., Stoff, H., and Wiedermann, A., 2009, "Effect of Wakes and Secondary Flow on Re-Attachment of Turbine Exit Annular Diffuser Flow," ASME J. Turbomach., **131**(4), p. 041012.
[83] Sieker, O., and Seume, J. R., 2008, "Influence of Rotating Wakes on Separation in Turbine Exhaust Diffusers," J. Therm. Sci., **17**(1), pp. 42–49.
[84] Mimic, D., Drechsel, B., and Herbst, F., 2018, "Correlation Between Pressure Recovery of Highly Loaded Annular Diffusers and Integral Stage Design Parameters1," ASME J. Turbomach., **140**(7), p. 071002.
[85] Weigand, B., and Spring, S., 2011, "Multiple Jet Impingement—A Review," Heat Trans. Res., **42**(2), pp. 101–142.

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[86] Zuckerman, N., and Lior, N., 2006, "Jet Impingement Heat Transfer: Physics, Correlations, and Numerical Modeling," *Advances in Heat Transfer*, 1st ed., Vol. 39, Elsevier Textbooks, United States, pp. 565–631.
[87] Sahiti, N., Lemouedda, A., Stojkovic, D., Durst, F., and Franz, E., 2006, "Performance Comparison of Pin Fin In-Duct Flow Arrays With Various Pin Cross-Sections," Appl. Therm. Eng., **26**(11–12), pp. 1176–1192.
[88] Armstrong, J., and Winstanley, D., 1987, "A Review of Staggered Array Pin Fin Heat Transfer for Turbine Cooling Applications," ASME. J. Turbomach., pp. 94–103.
[89] Han, J.-C., Dutta, S., and Ekkad, S., 2013, *Gas Turbine Heat Transfer and Cooling Technology*, 2nd ed., CRC Press, Boca Raton, FL.
[90] Chen, A. F., Shiau, C.-C., Han, J.-C., and Krewinkel, R., 2019, "Heat Transfer in a Rotating Two-Pass Rectangular Channel Featuring a Converging Tip Turn With Various 45 deg Rib Coverage Designs," ASME J. Therm. Sci. Eng. Appl., **11**(6), p. 061015.
[91] Chen, I.-L., Sahin, I., Wright, L. M., Han, J.-C., and Krewinkel, R., 2022, "Heat Transfer in a Rotating, Blade-Shaped, Two-Pass Cooling Channel With Various 45-Deg Rib Orientations," ASME J. Therm. Sci. Eng. Appl., **14**(9), p. 091009.
[92] Bogard, D. G., and Thole, K. A., 2006, "Gas Turbine Film Cooling," J. Propul. Power, **22**(2), pp. 249–270.
[93] Takeishi, K., 2022, "Evolution of Turbine Cooled Vanes and Blades Applied for Large Industrial Gas Turbines and Its Trend Toward Carbon Neutrality," Energies, **15**(23), p. 8935.
[94] Town, J., Straub, D., Black, J., Thole, K. A., and Shih, T. I.-P., 2018, "State-of-the-Art Cooling Technology for a Turbine Rotor Blade," ASME J. Turbomach., **140**(7), p. 071007.
[95] Landfester, C., Müller, G., Krewinkel, R., Domnick, C., and Böhle, M., 2022, "Comparison of Film Cooling Performance for Different Purge Slot Configurations in a Cylindrical and State-of-the-Art Nozzle Guide Vane," ASME J. Turbomach., **144**(3), p. 031014.
[96] Friedrichs, S., Hodson, H. P., and Dawes, W. N., 1999, "The Design of an Improved Endwall Film-Cooling Configuration," ASME J. Turbomach., **121**(4), pp. 772–780.
[97] Takeishi, K., Oda, Y., and Kozono, S., 2015, "Experimental Study of Leakage Flow on Flow Field and Film Cooling of High Pressure Turbine Blade Platform," Proceedings of the ASME Turbo Expo: Turbine Technical Conference and Exposition—2015, ASME Turbo Expo 2015: Turbine Technical Conference and Exposition, Vol. 5, ASME, Montreal, QC, Canada, June 15–19, 2015.
[98] Hada, S., and Thole, K. A., 2011, "Computational Study of a Midpassage Gap and Upstream Slot on Vane Endwall Film-Cooling," ASME J. Turbomach., **133**(1), p. 011024.
[99] Thomas, M., and Povey, T., 2017, "Improving Turbine Endwall Cooling Uniformity by Controlling Near-Wall Secondary Flows," Proc. Inst. Mech. Eng. Part G J. Aerosp. Eng., **231**(14), pp. 2689–2705.
[100] Landfester, C., Krewinkel, R., and Böhle, M., eds., 2024, "Experimentelle Untersuchung von Plattform-Kühlung in Einer Gekrümmten Leitschaufel-Kaskade," 18th Status Seminar, Cologne, Germany, Apr. 4–9, 2024.
[101] Nealy, D. A., and Reider, S. B., 1980, "Evaluation of Laminated Porous Wall Materials for Combustor Liner Cooling," ASME J. Eng. Gas Turbines Power, **102**(2), pp. 268–276.
[102] Krewinkel, R., 2013, "A Review of Gas Turbine Effusion Cooling Studies," Int. J. Heat Mass Transfer, **66**, pp. 706–722.
[103] Al Dabagh, A. M., Andrews, G. E., Abdul Husain, R. A. A., Husain, C. I., Nazari, A., and Wu, J., 1990, "Impingement/Effusion Cooling: The Influence of the Number of Impingement Holes and Pressure Loss on the Heat Transfer Coefficient," ASME J. Turbomach, **112**(3), pp. 467–476.
[104] Courtis, M., Murray, A., Coulton, B., Ireland, P., and Mayo, I., 2021, "Influence of Spanwise and Streamwise Film Hole Spacing on Adiabatic Film Effectiveness for Effusion-Cooled Gas Turbine Blades," IJTPP, **6**(3), p. 37.
[105] Straub, D., Ramesh, S., Searle, M., Robey, E., Floyd, T., Roy, A., and Ames, F. E., 2022, "Performance of Additively Manufactured Internally Cooled Airfoils for Small Industrial Gas Turbines," Proceedings of ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition (GT 2022), ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition, The American Society of Mechanical Engineers, Rotterdam, The Netherlands, June 13–17.
[106] Petrovic, M., Petkovic, D., Milic, S., Wiedermann, A., and Krewinkel, R., 2025, "Comprehensive Method for Predicting Gas Turbine Cycle Performances Considering the Impact of Various Fuels," ASME J. Eng. Gas Turbines Power, pp. 1–36. (Paper No: GTP-25-1359, ASME Digital Collection).
[107] Ling, J., Coletti, F., Yapa, S. D., and Eaton, J. K., 2013, "Experimentally Informed Optimization of Turbulent Diffusivity for a Discrete Hole Film Cooling Geometry," Int. J. Heat Fluid Flow, **44**, pp. 348–357.
[108] Milani, P. M., Ling, J., Saez-Mischlich, G., Bodart, J., and Eaton, J. K., 2018, "A Machine Learning Approach for Determining the Turbulent Diffusivity in Film Cooling Flows," ASME J. Turbomach., **140**(2), p. 021006.
[109] Misaka, T., Asaumi, N., Kubo, S., Ideta, T., and Obayashi, S., 2019, "Prediction of Film-Cooling Effectiveness Based on Bayesian Model Calibration of SST Turbulence Model," IGTC-2019-061.
[110] Sandberg, R. D., Tan, R., Weatheritt, J., Ooi, A., Haghiri, A., Michelassi, V., and Laskowski, G., 2018, "Applying Machine Learnt Explicit Algebraic Stress and Scalar Flux Models to a Fundamental Trailing Edge Slot," ASME J. Turbomach., **140**(10), p. 101008.
[111] Bischoff, T., Voigt, M., Chehab, E., and Vogeler, K., 2006, "Probabilistic Analysis of Stationary Gas Turbine Secondary Air Systems," Proceedings of the ASME Turbo Expo 2006, ASME Turbo Expo 2006: Power for Land, Sea, and Air, Vol. 3, ASME, Barcelona, Spain, May 8–11, 2006, pp. 1365–1373.
[112] Nasti, A., Voutchkov, I. I., Toal, D. J. J., and Keane, A. J., 2022, "Multi-fidelity Simulation for Secondary Air System Seal Design in Aero Engines," Proceedings of ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition (GT 2022), ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition, The American Society of Mechanical Engineers, Rotterdam, The Netherlands, June 13–17, 2022.
[113] Uyav, O., and Bedir, H., 2023, "Investigation of Power Turbine Casing Thermal Environment Using One Dimensional Thermal-Fluid Network Solver," Proceedings of ASME Turbo Expo 2023: Turbomachinery Technical Conference and Exposition (GT 2023), ASME Turbo Expo 2023: Turbomachinery Technical Conference and Exposition, The American Society of Mechanical Engineers, Boston, MA, June 26–30, 2023.
[114] Khosravy, M., 2013, *Progress in Gas Turbine Performance*, IntechOpen, United States.
[115] Molière, M., 2023, "The Fuel Flexibility of Gas Turbines: A Review and Retrospective Outlook," Energies, **16**(9), p. 3962.
[116] Stefanizzi, M., Capurso, T., Filomeno, G., Torresi, M., and Pascazio, G., 2021, "Recent Combustion Strategies in Gas Turbines for Propulsion and Power Generation Toward a Zero-Emissions Future: Fuels, Burners, and Combustion Techniques," Energies, **14**(20), p. 6694.
[117] Becker, C. M., Marder, M., Junges, E., and Konrad, O., 2022, "Technologies for Biogas Desulfurization—An Overview of Recent Studies," Renew. Sustain. Energy Rev., **159**, p. 112205.
[118] EUGINE—European Engine Power Plants Association. "Hydrogen-Ready Engine Power Plants," https://www.eugine.eu/h2-ready/new-plants/. Accessed October 28, 2024.
[119] Öberg, S., Odenberger, M., and Johnsson, F., 2022, "The Value of Flexible Fuel Mixing in Hydrogen-Fueled Gas Turbines—A Techno-Economic Study," Int. J. Hydrogen Energy, **47**(74), pp. 31684–31702.
[120] Bexten, T., Wirsum, M., Roscher, B., Schelenz, R., Jacobs, G., Weintraub, D., and Jeschke, P., 2019, "Optimal Operation of a Gas Turbine Cogeneration Unit With Energy Storage for Wind Power System Integration," ASME J. Eng. Gas Turbines Power, **141**(1), p. 011801.
[121] Robb, D. Fuel Switching—Gas Turbines Running on Hydrogen or a Natural Gas and Hydrogen Mixture could Aid in Emissions Reduction." https://www.turbomachinerymag.com/view/fuel-switching, Accessed October 25, 2024.
[122] Lam, K.-K., Geipel, P., and Larfeldt, J., 2015, "Hydrogen Enriched Combustion Testing of Siemens Industrial SGT-400 at Atmospheric Conditions," ASME J. Eng. Gas Turbines Power, **137**(2), p. 021502.
[123] Mitsubishi Power. "Turbines Driven Purely by Hydrogen in the Pipeline," https://www.nature.com/articles/d42473-020-00545-7, Accessed October 25, 2024.
[124] Horikawa, A., Okada, K., Yamaguchi, M., Aoki, S., Wirsum, M., Funke, H. H.-W., and Kusterer, K., 2021, "Combustor Development and Engine Demonstration of Micro-Mix Hydrogen Combustion Applied to M1A-17 Gas Turbine," Proceedings of ASME Turbo Expo 2021: Turbomachinery Technical Conference and Exposition, ASME Turbo Expo 2021: Turbomachinery Technical Conference and Exposition, Virtual, Online, The American Society of Mechanical Engineers, June 7–11.
[125] Ćosić, B., Wassmer, D., Kluß, D., Jaeschke, A., Reichel, T., and Paschereit, C. O., 2022, "Experimental and Numerical Advancement of the MGT Combustor Towards Higher Hydrogen Capabilities," Volume 3B: Combustion, Fuels, and Emissions, ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition, American Society of Mechanical Engineers, Rotterdam, The Netherlands, June 13–17.
[126] Prandi, R., 2021, "A Hotter Future for Gas Turbines?," Diesel & Gas Turbine Worldwide. https://www.powerprogress.com/magazines/diesel-gas-turbine-worldwide-autumn-2021/8015353.article, Accessed October 25, 2024.
[127] "HYdrogen as a FLEXible Energy Storage for a Fully Renewable European POWER system," EC HORIZON 2020 Project ID 884229, Closed in 2024. .
[128] "Flexibility for Hydrogen," EC HORIZON EUROPE Project ID 101101427, Ongoing, Closing Date 2026.
[129] Bellotti, D., Meloni, R., and Pucci, E., 2023, "Ammonia as Fuel for Gas Turbine," E3S Web of Conferences, 2007, Savona, Italy, Sept. 6–8, p. 414.
[130] Kobayashi, H., Hayakawa, A., Somarathne, K. K. A., and Okafor, E. C., 2019, "Science and Technology of Ammonia Combustion," Proc. Combust. Inst., **37**(1), pp. 109–133.
[131] Kurata, O., Iki, N., Fan, Y., Matsunuma, T., Inoue, T., Tsujimura, T., Furutani, H., et al., 2021, "Start-Up Process of 50kW-Class Gas Turbine Firing Ammonia Gas," Turbo Expo: Power for Land, Sea, and Air, Vol. 84997, p. V006T19A012.
[132] Bonasio, V., and Ravelli, S., 2022, "Performance Analysis of an Ammonia-Fueled Micro Gas Turbine," Energies, **15**(11), pp. 3874.
[133] Verkamp, F. J., Hardin, M. C., and Williams, J. R., 1967, "Ammonia Combustion Properties and Performance in Gas-Turbine Burners," Symp. (Int.) Combust., **11**(1), pp. 985–992.
[134] Nose, N., Kawakami, T., Nakamura, S., Kuroki, H., Kataoka, M., and Yuri, M., 2021, "Development of Hydrogen/Ammonia Firing Gas Turbine for Decarbonized Society," MHI Tech. Rev., **58**(3), pp. 1–11.
[135] Ito, S., Uchida, M., Suda, T., and Fujimori, T., 2023, "Demonstration Tests of Ammonia/Natural Gas Co-Firing Power Generation With a 2-MW-Class Gas Turbine," CO_2 Free Ammonia as an Energy Carrier, Japan's Insights, 1st ed., Springer Nature Singapore, Singapore, pp. 515–522.

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[136] Langella, G., Sorrentino, G., Sabia, P., Ariemma, G. B., Amoresano, A., and Iodice, P., 2023, "Ammonia as a Fuel for Gas Turbines: Perspectives and Challenges," J. Phys., **2648**(1), p. 12009.
[137] Park, S., Shin, J., Park, J., Lee, S., and Choi, N., 2023, "Comparison of Hydrogen and Ammonia Co-Firing With Natural Gas Using a Practical Gas Turbine Combustor (501F) Under Atmospheric Conditions: Changes in Metal Temperature, Pattern Factor, and NOx Emission," Turbo Expo: Power for Land, Sea, and Air, Boston, MA, June 26–30.
[138] Nose, M., Araki, H., Senba, N., Furuichi, H., and Tanimura, S., 2022, "Development of Ammonia Utilization Technology for Large Gas Turbines for Power Generation," CO_2 Free Ammonia as an Energy Carrier: Japan's Insights, Springer Nature Singapore, Singapore, pp. 523–535.
[139] Wiedermann, A., Ladwig, M., Bergins, C., Bernstrauch, O., Herce-Fuente, C., and Jansohn, P. Flexible Power Generation in a Decentralized Europe," ETIP-SNET White Paper. https://smart-networks-energy-transition.ec.europa.eu/sites/default/files/publications/WG3_WhitePaper_Flexible-Power-Generation-in-a-Decarbonised-Europe.pdf, Accessed October 25, 2024.
[140] Iki, N., 2022, "Demonstration of Ammonia Fuel With Small Gas Turbines," CO_2 Free Ammonia as an Energy Carrier Japan's Insights, Springer Nature Singapore, Singapore, pp. 487–499.
[141] Fawwaz Alrebei, O., Hamdan Al Assaf, A., Al-Kuwari, M. S., and Amhamed, A., 2022, "Lightweight Methane-Air Gas Turbine Controller and Simulator," Energy Convers. Manage:X, **15**, p. 100242.
[142] Pashchenko, D., Mustafin, R., and Karpilov, I., 2022, "Efficiency of Chemically Recuperated Gas Turbine Fired With Methane: Effect of Operating Parameters," Appl. Therm. Eng., **212**, p. 118578.
[143] Pashchenko, D., Mustafin, R., and Karpilov, I., 2022, "Thermochemical Recuperation by Steam Methane Reforming as an Efficient Alternative to Steam Injection in the Gas Turbines," Energy, **258**, p. 124913.
[144] Mevissen, F., and Meo, M., 2020, "A Nonlinear Ultrasonic Modulation Method for Crack Detection in Turbine Blades," Aerospace, **7**(6), p. 72.
[145] Mevissen, F., and Meo, M., 2022, "Ultrasonically Stimulated Thermography for Crack Detection of Turbine Blades," Infrared Phys. Technol., **122**, p. 104061.
[146] Mevissen, F., and Meo, M., 2021, "Nonlinear Ultrasound Crack Detection With Multi-frequency Excitation—A Comparison," Sensors, **21**(16), p. 5368.
[147] Jambol, D. D., Sofoluwe, O. O., Ukato, A., and Ochulor, O. J., 2024, "Transforming Equipment Management in Oil and Gas With AI-Driven Predictive Maintenance," Comput. Sci. IT Res. J., **5**(5), pp. 1090–1112.
[148] Fahmi, A. W., Kashyzadeh, K. R., and Ghorbani, S., 2024, "Advancements in Gas Turbine Fault Detection: A Machine Learning Approach Based on the Temporal Convolutional Network–Autoencoder Model," Appl. Sci., **14**(11), p. 4551.
[149] Zhu, H., Dai, S., Zhang, X., Chen, J., Luo, M., and Huang, W., 2023, "An Online Fatigue Damage Evaluation Method for Gas Turbine Hot Components," Energies, **16**(19), p. 6785.
[150] Zhu, H., Zhu, Y., Zhang, X., Chen, J., Luo, M., and Huang, W., 2023, "Research on a Method for Online Damage Evaluation of Turbine Blades in a Gas Turbine Based on Operating Conditions," Aerospace, **10**(11), p. 966.
[151] Han, Z., Jiang, Z., Ehrt, S., and Li, M., 2021, "Low-Cycle Fatigue Lifetime Estimation and Predictive Maintenance for a Gas Turbine Compressor Vane Carrier Under Varying Operating Conditions," ASME J. Mech. Des., **143**(7), p. 072003.
[152] Hu, M., He, Y., Lin, X., Lu, Z., Jiang, Z., and Ma, B., 2023, "Digital Twin Model of Gas Turbine and Its Application in Warning of Performance Fault," CJA, **36**(3), pp. 449–470.
[153] Cruz-Manzo, S., Panov, V., and Bingham, C., 2023, "GAS Turbine Sensor Fault Diagnostic System in a Real-Time Executable Digital-Twin," J. Glob. Power Propuls. Soc., **7**, pp. 85–94.
[154] Apostolidis, A., Bouriquet, N., and Stamoulis, K. P., 2022, "AI-Based Exhaust Gas Temperature Prediction for Trustworthy Safety-Critical Applications," Aerospace, **9**(11), p. 722.
[155] Gotoh, J., Hayasaka, Y., Sakurai, S., and Ikeda, H., 2002, "Remote Monitoring Life Management System for Gas Turbines," JSME Int. J. Ser. B Fluid. Therm. Eng., **45**(3), pp. 536–542.
[156] Petkovic, D., Banjac, M., Milic, S., Petrovic, M. V., and Wiedermann, A., 2020, "Modeling the Transient Behavior of Gas Turbines," ASME J. Turbomach., **142**(8), p. 081005.
[157] Turbomachinery for Green Fuels and Energy Storage (TurboGruen). "Funding Contract No. 5EE5068A, Federal Ministry for Economic Affairs and Climate, Actions, Berlin, Germany." https://www.energieforschung.de/en/home/news/2022/turbomachinery-for-green-fuels-and-energy-storage, Accessed April 1, 2025.
[158] Rustamov, N., Meirbekova, O., Kibishov, A, Babakhan, S., and Berguzinov, A, 2022, "Creation of a Hybrid Power Plant Operating on the Basis of a Gas Turbine Engine," EEJET, **2**(8 (116)), pp. 29–37.
[159] Ancona, M. A., Bianchi, M., Branchini, L., Pascale, A. D., Melino, F., Peretto, A., and Torricelli, N., 2021, "Systematic Comparison of ORC and s-CO2 Combined Heat and Power Plants for Energy Harvesting in Industrial Gas Turbines," Energies, **14**(12), p. 3402.
[160] Pilavachi, P., 2000, "Power Generation With Gas Turbine Systems and Combined Heat and Power," Appl. Therm. Eng., **20**(15–16), pp. 1421–1429.
[161] Bespalov, V. I., Gurova, O. S., Lysova, E. P., and Grishin, G. S., 2023, "Life Cycle Analysis of Steam and Gas Turbine CHP Plants," Modern Trend. Constr. Urban Territorial Plann., **1**(4), pp. 32–43.
[162] Cormos, C.-C., Cormos, A.-M., and Agachi, S., 2009, "Heat and Power Integration for Hydrogen-Fuelled Combined Cycle Gas Turbine (CCGT)," 19th European Symposium on Computer Aided Process Engineering, Cracow, Poland, June 14–17, Vol. 26, Elsevier, pp. 1239–1244.
[163] Komarov, I., Zlyvko, S., Vegera, O., and Naumov, V., 2021, "Combined Cycle Gas Turbine for Combined Heat and Power Production With Energy Storage by Steam Methane Reforming," J. Energy Syst., **5**(3), pp. 231–243.
[164] Pantaleo, A. M., Camporeale, S., and Fortunato, B., 2015, "Small Scale Biomass CHP: Techno-Economic Performance of Steam Vs Gas Turbines With Bottoming ORC," Energy Proc., **82**, pp. 825–832.
[165] Bălănescu, D.-T., and Homutescu, V.-M., 2019, "Performance Analysis of a Gas Turbine Combined Cycle Power Plant With Waste Heat Recovery in Organic Rankine Cycle," Proc. Manuf., **32**, pp. 520–528.
[166] Sabzpoushan, S., Morad, M. R., and Ebrahimi Rahnama, H., 2022, "A Combined Cooling and Power Transcritical CO2 Cycle for Waste Heat Recovery From Gas Turbines," TSEP, **34**, p. 101423.
[167] Suomalainen, M. S., Arasto, A., Teir, S., and Siitonen, S., 2013, "Improving a Pre-Combustion CCS Concept in Gas Turbine Combined Cycle for CHP Production," Energy Proc., **37**, pp. 2327–2340.
[168] Liu, X., Fei, C., Yao, Y., Aladdin, M., and Su, Z., 2024, "Energy, and Environmental Investigation of a Hybrid Gas Turbine–Solar Energy for Desalination Process for Using in Sport Stadiums," Int. J. Low-Carbon Technol., **19**, pp. 33–42.
[169] Moreno-Gamboa, F., Guerrero-Gomez, G., and Caicedo-Rolon, A., 2023, "Energy Losses Study in Solar Hybrid Gas Turbine in Colombia," EUREKA Phys. Eng., **29**(5), pp. 35–45.
[170] Sreekanth, M., and Daniel, J., 2016, "Exergy in Hybrid Gas Turbines- A Review," Indian J. Sci. Technol., **9**(40), pp. 1–10.
[171] Behar, O., Belgasim, B., Sbarbaro, D., and Moran, L., 2021, "Influence of the Geographical Parameters on the Performance of Hybrid Solar Gas Turbine," Int. J. Energetica, **6**(1), pp. 28–34.
[172] Fenner, C., and van der Spuy, S. J., 2022, "Improved Modeling and Experimental Evaluation of a Micro Gas Turbine for Solar-Hybrid Application," Proceedings of ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition (GT 2022), ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition, The American Society of Mechanical Engineers, Rotterdam, The Netherlands, June 13–17.
[173] Gunaltili, E., Ekici, S., Yilmazoglu, M. Z., and Karakoc, T. H., 2023, "Gas Turbine and Fuel Cell Hybrid Systems," Emerging Trends in Electric Aviation, Proceedings of the International Symposium on Electric Aviation and Autonomous Systems 2022, Sustainable Aviation, Warsaw, Poland, July 5–7, Springer Nature Switzerland; Imprint Springer, pp. 1–7.
[174] Jia, K., Liu, C., Li, S., and Jiang, D., 2023, "Modeling and Optimization of a Hybrid Renewable Energy System Integrated With Gas Turbine and Energy Storage," Energy Convers. Manage., **279**, p. 116763.
[175] Patel, S. J., Wu, D., McKinney, R., Emerson, B. L., and Lieuwen, T. C., 2022, "Development of an Experimental Combustor for Hybrid Electric Gas Turbines," GPPS Chania22, Proceedings of Global Power & Propulsion Society (GPPS), Chania, Crete, Greece, Sept. 12–14.
[176] Rosner, F., and Samuelsen, S., 2022, "Thermo-Economic Analysis of a Solid Oxide Fuel Cell-Gas Turbine Hybrid With Commercial Off-the-Shelf Gas Turbine," Appl. Energy, **324**, p. 119745.
[177] Seitz, A., Nickl, M., Troeltsch, F., and Ebner, K., 2022, "Initial Assessment of a Fuel Cell—Gas Turbine Hybrid Propulsion Concept," Aerospace, **9**(2), p. 68.
[178] Sinha, A. A., Choudhary, T., Ansari, M. Z., Shukla, A. K., and Arabkoohsar, A., 2023, "Waste Heat Recovery and Exergy-Based Comparison of a Conventional and a Novel Fuel Cell Integrated Gas Turbine Hybrid Configuration," Sustain. Energy Technol. Assess., **57**, p. 103256.
[179] Subotić, V., and Hochenauer, C., 2022, "Analysis of Solid Oxide Fuel and Electrolysis Cells Operated in a Real-System Environment: State-of-the-Health Diagnostic, Failure Modes, Degradation Mitigation and Performance Regeneration," Prog. Energy Combust. Sci., **93**, pp. 101011.
[180] Fortune Business Insights. "Gas Turbine Market Size." https://www.fortunebusinessinsights.com/gas-turbine-market-106255, Accessed August 14, 2024.
[181] Grand View Research. "Gas Turbine Market Size & Trends." https://www.grandviewresearch.com/industry-analysis/gas-turbine-market, Accessed August 14, 2024
[182] Global Market Insights. "Industrial Gas Turbine Market Size." https://www.gminsights.com/industry-analysis/industrial-gas-turbine-market, Accessed August 14, 2024.
[183] MarketsandMarkets Research. "Gas Turbine Markets." https://www.marketsandmarkets.com/Market-Reports/gas-turbines-market-94641697.html, Accessed August 14, 2024.
[184] ETN Global, 2024, "Hydrogen Gas Turbines." https://etn.global/, Accessed December 9, 2024
[185] IEA, 2022, "Global Hydrogen Review 2022." https://www.iea.org/reports/global-hydrogen-review-2022, Accessed October 28, 2024.
[186] Powertheeu. "EU Turbines." https://powertheeu.eu, Accessed October 28, 2024.
[187] Lee, S., Kim, H. S., Park, J., Kang, B. M., Cho, C.-H., Lim, H., and Won, W., 2021, "Scenario-Based Techno-Economic Analysis of Steam Methane Reforming Process for Hydrogen Production," Appl. Sci., **11**(13), p. 6021.
[188] Esteban, M., and Romeo, L. M., 2021, "Techno-Economics Optimization of H2 and CO2 Compression for Renewable Energy Storage and Power-to-Gas Applications," Appl. Sci., **11**(22), p. 10741.
[189] Fortune Business Insights. "Gas Turbine Market Size." https://www.fortunebusinessinsights.com/gas-turbine-market-106255, Accessed July 11, 2024.

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[190] Power Technology. "Magnum Power Plant, The Netherlands," Plans for the Magnum Power Plant Would See It Converted Into a Zero-Carbon Power Generation Facility Using Hydrogen for Fuel. https://www.power-technology.com/projects/nuonmagnum-igcc/?cf-view, Accessed February 26, 2025.
[191] BloombergNEF. "Green Hydrogen to Undercut Gray Sibling by End of Decade." https://about.bnef.com/insights/clean-energy/green-hydrogen-to-undercut-gray-sibling-by-end-of-decade/, Accessed September 19, 2025.
[192] IEA, 2024, "Global Hydrogen Review." https://www.iea.org, Accessed December 9, 2024.
[193] Coresite. "Breaking Down Data Center Tier Level Classifications." https://www.coresite.com/blog/breaking-down-data-center-tiers-classifications, Accessed September 17, 2025.
[194] Chevron. "Engine No. 1, Chevron and GE Vernova to Power U.S. Data Centers." https://www.chevron.com/newsroom/2025/q1/power-solutions-for-us-data-centers, Accessed September 17, 2025.
[195] Siemens Energy. "Eaton and Siemens Energy Join Forces to Provide Power and Technology to Accelerate the Delivery of New Data Center Capacity." https://www.siemens-energy.com/us/en/home/press-releases/eaton-and-siemens-energy-join-forces-to-provide-power-and-techno.html, Accessed September 17, 2025.
[196] Fortune Business Insights. "Data Center Market." https://www.fortunebusinessinsights.com/data-center-market-109851, Accessed September 17, 2025.
[197] PwC. "Data Centers at the Crossroads of Technology and Resilience." https://www.pwc.com/us/en/industries/tmt/library/hyperscale-data-center.html, Accessed September 17, 2025.
[198] Grand View Research. "AI Data Center Market Summary." https://www.grandviewresearch.com/industry-analysis/ai-data-center-market-report, Accessed September 17, 2025.
[199] Zero Emissions Platform (ZEP). "The Costs of CO2 Capture, Transport and Storage." www.zeroemissionsplatform.eu, Accessed October 28, 2024.
[200] Nagashima, M., 2018, *Japan's Hydrogen Strategy and Its Economic and Geopolitical Implications*, Etudes de l'Ifri, IFRI, Paris, France.
[201] KIZUNA. Linking Japan and the World "Clean Energy Strategy to Achieve Carbon Neutrality by 2050." https://www.japan.go.jp/kizuna/2022/06/clean_energy_strategy.html, Accessed October 28, 2024.
[202] Kawasaki Heavy Industries. "The Pathway to Hydrogen," Paving the Way for a Hydrogen-Based Society. https://global.kawasaki.com/en/hydrogen/, Accessed October 28, 2024.
[203] ASUE. "Entwicklung der Low-NOx Micro-Mix H2- Brennkammer für Industriegasturbinen. Tagungsband zum ASUE Expertenkreis Gasturbinentechnik." https://www.asue.de/sites/default/files/asue/themen/gasturbinen/2018/broschueren/ASUE_Tagungsband-EK-Gasturbinentechnik_Muenchen-2018.pdf, Accessed October 28, 2024.
[204] COMPRESSORTECH2, 2024, COMPRESSORTECH2. https://www.compressortech2.com/magazines/COMPRESSORTECH2-December-2018/1136119.article, Accessed October 28, 2024.
[205] MHI. "PTT and Mitsubishi Heavy Industries Asia Pacific to Explore 100% Ammonia Gas Turbine Powered Generation in Thailand," MHI Press Release. https://www.mhi.com/news/24030801.html, Accessed February 26, 2025.
[206] Robb, D. "Fuel Switching," Gas Turbines Running on Hydrogen or a Natural Gas and Hydrogen Mixture Could Aid in Emissions Reduction. Turbomachinery International. https://www.turbomachinerymag.com/view/fuel-switching, Accessed October 28, 2024.

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